

Implementation details

Datasets

We conducted experiments on the following real-world datasets to evaluate the performance of the proposed method. The datasets include:

- **Weather:** 21 meteorological factors were collected every 10 minutes from the Weather Station of the Max Planck Biogeochemistry Institute in 2020.
- **Traffic:** Hourly road occupancy rates were recorded from 862 sensors installed on freeways in the San Francisco Bay Area, covering the period from January 2015 to December 2016.
- **Electricity:** Hourly electricity consumption data from 321 clients were recorded.
- **Solar:** Solar power production data from 137 photovoltaic plants in 2006 were collected every 10 minutes.
- **ETT:** Data on 7 factors of electricity transformers were collected from July 2016 to July 2018. Four subsets are available: ETTh1 and ETTh2, recorded every hour, and ETTm1 and ETTm2, recorded every 15 minutes.
- **PEMS:** Public traffic network data from California were sampled in 5-minute windows.

For more information about datasets, please refer to Liu et al. (2024).

Base point prediction models

We did not conduct hyperparameter search suggested by the original papers (Liu et al. 2024; Han et al. 2024; Yu et al. 2024) because finding the best point prediction model is not our focus. As a result, we fix the dimension of feature as 256 and the number of layer as 3. Besides, we used Adam to train these models for 50 epochs with early stop if loss in validation set did not decrease for 5 epochs. The learning rate was 0.001. Other hyperparameters were set as default values according to their github repositories.

Quantile prediction models

For our own method, MLP was considered as quantile prediction model. The learning rate γ is a critical parameter. In the main experiments, we fixed γ at 0.002. We set two hidden layers in MLP model with hidden size 512 and 256. We split the validation set randomly by 8:2, and used 80% of data to train the quantile prediction model for 100 epochs. If loss in the remaining 20% does not decrease for 5 epochs, we would stop training. Adam was used as optimization with learning rate 0.001. We set α as 0.1, which means we want to obtain confidence interval with 90% coverage.

Resource requirements

All our experiments were conducted on a single RTX 4090 GPU. In terms of computational resource requirements, training the quantile regression model using features consumes approximately 2GB of GPU memory. The training time varies depending on the dataset size—for the ETT dataset, it takes only about 2 minutes, while for larger datasets such as PEMS07 and the Traffic dataset, training requires around 30 minutes.

Related Work

Conformal prediction and its single variable time series version

The simplest form of conformal prediction treats the errors on the validation set as representative of those on the test set (Shafer and Vovk 2008). Specifically, if the 90th percentile of the validation errors is q , and $\hat{y}$ is a point prediction on the test set, then the corresponding 90% confidence interval is given by $[\hat{y} - q, \hat{y} + q]$. Furthermore, under the assumption of exchangeability, Lei et al. (2018) proved that such confidence intervals achieve the desired coverage level with theoretical guarantees.

However, for time series data, the values at adjacent time points are usually dependent, and the distribution of data may also change over time. Therefore, considerable research has been done to improve the basic conformal prediction method to develop algorithms for these non-exchangeable time series data. One foundational work in this area is ACI (Gibbs and Candes 2021). Its core idea is to adjust the confidence interval during deployment. If the coverage rate of the previous step is too large, the confidence interval will be shortened, and vice versa. Later, some works committed to improve ACI, specifically focusing on how to determine the adjusting speed of interval (Xu, Jiang, and Xie 2024). They treated the problem as an online convex optimization problem, and provided algorithms to form this framework (Bhatnagar et al. 2023; Zhang, Lu, and Yang 2024; Zhang, Bombara, and Yang 2024; Angelopoulos, Barber, and Bates 2024). Additionally, some studies proposed methods to construct confidence intervals by continuously updating validation set (Xu and Xie 2021) (EnbPI) or giving distinctive weight to each data point in validation set (Barber et al. 2023) or by predicting future errors using the errors from some past time steps (SPCI) (Xu and Xie 2023).

Conformal prediction for multi-variable or multi-step time series

In recent work about multi-step time series prediction (Sun and Yu 2022), different steps were just regarded as different variables, so we list works about multi-variable and multi-step as a single class and use variable to represent both step and variable in the following part of this section. In terms of constructing confidence intervals for multivariate time series, there are two types of coverage requirement.

Suppose that the prediction is of dimension p . The first type treats the p -dimensional data as a single data point and then constructs a region in the p -dimensional space such that the true value falls within this region with a given probability (Xu, Jiang, and Xie 2024; Stankeviciute, M Alaa, and van der Schaar 2021). The core issue here is how to consider the relationships between different dimensions. Some methods enlarged the given confidence level $1 - \alpha$ (Stankeviciute, M Alaa, and van der Schaar 2021; Cleaveland et al. 2024; Lopes et al. 2024), while others used copula methods (Sun and Yu 2022). Besides, some researchers considered the covariance of errors in different dimensions and then constructed ellipsoidal prediction regions (Xu, Jiang, and Xie 2024). The second type treats the data as p data points, and provides a confidence interval for each dimension individually, requiring each dimension to fall within its interval with a given probability. Some methods extended the ACI method to high-dimensional data (Lin, Trivedi, and Sun 2022; Yang, Candès, and Lei 2024), for example, applied ACI in each dimension. And some work also considered using past errors to predict future errors (Batra, Mercuri, and Khraishi 2023), similar to SPCI. Besides, EnbPI (Xu and Xie 2021) and ACI (Gibbs and Candès 2021) have been combined in (Sousa, Tomé, and Moreira 2024).

Better intervals for conformal prediction

Another category of methods relevant to our research are how to construct more efficient conformal prediction intervals. The main issue with traditional conformal prediction is that it gives confidence interval with the same length for every point, which is not suitable for many problems. Therefore, some studies proposed to construct confidence intervals based on local sample points (Lee, Xu, and Xie 2024; Lei and Wasserman 2014) or use the predicted variance to construct the confidence intervals (Lei et al. 2018). Other works transformed traditional predictive models into quantile regression models and adjusting the predicted quantiles using validation set (Romano, Patterson, and Candès 2019). Additionally, some researchers considered modifying point prediction models to distribution prediction models and then construct confidence intervals (Chernozhukov, Wüthrich, and Zhu 2021). In recent studies, the features of data have been considered, for example, by dividing the feature space into different regions and providing a distinctive confidence interval length for each region (Kiyani, Pappas, and Hassani 2024a,b). Some papers also define errors at the feature level and then use feature-level errors to derive errors at the prediction level (Teng et al. 2023; Chen, Ren, and Cheng 2024).

Baseline methods

In this section, we first introduce each baseline method and then provide some discussion on several baselines most relevant to our approach.

ECI (Wu et al. 2025)

This method employs a smoother loss function as an alternative to the traditional quantile loss used in online conformal prediction. Specifically, the original quantile loss function (Equation 2) contains an indicator function that determines whether s_t exceeds q_t . The authors argue that this indicator function is excessively non-smooth and propose replacing it with a sigmoid function. Consequently, they derive the following updated formulation for q_t :

$$q_{t+1} = q_t + \gamma(err_t - \alpha - (s_t - q_t)\nabla f(s_t - q_t)) \quad (16)$$

where $err_t = \mathbb{I}(s_t > q_t)$, f means sigmoid function, i.e., $f(x) = \frac{1}{1 + \exp(-x)}$. Although the original paper used $c = 1$ in their experiments, we found this setting failed to achieve the target coverage rate in our experiments. Through empirical testing, we determined that $c = 0.2$ yields better results. Consequently, all reported results in our study correspond to this configuration.

TQA-E (Lin, Trivedi, and Sun 2022)

This method is designed to tackle the multi-variable problem and we modified it in our experiments for multi-step problem. We use the $1 - a_{t,i,j}$ quantile of errors in the past to obtain confidence intervals. That is,

$$C_{t,i,j} = [\hat{y}_{t,i,j} - Q_{1-a_{t,i,j}}(E_{t-j,i,j}), \hat{y}_{t,i,j} + Q_{1-a_{t,i,j}}(E_{t-j,i,j})] \quad (17)$$

Where $E_{t,i,j} = \{e_{k,i,j} \mid k \in [0, 1, \dots, t]\}$ is a bag (Shafer and Vovk 2008) of errors in the past times. And $a_{t,i,j} = \alpha - \delta_{t,i,j}$, $\delta_{t,i,j}$ is obtained in an iteration way:

$$\delta_{t+1,i,j} = \begin{cases} \delta_{t,i,j} + \gamma(I_{y_{t-j,i,j} \notin C_{t-j,i,j}} - \alpha) & \delta_{t,i,j} \geq 1 - \alpha \\ (1 - \gamma)\delta_{t,i,j} & otherwise \end{cases} \quad (18)$$

γ was set as 0.005 according to the original paper.

LPCI (Batra, Mercuri, and Khraishi 2023)

The main idea of this method is using $[e_{t-k}, \dots, e_{t-1}]$ to predict the quantile of $e_t = y_t - \hat{y}$ and the confidence interval is

$$[\hat{y}_t + q_\beta(e_t), \hat{y}_t + q_{1-\alpha+\beta}(e_t)]$$

where β is chosen to minimize to interval length. To handle multi-variable situation, Batra, Mercuri, and Khraishi (2023) proposes to use data in all dimensions to fit one model but using additional variable identifies ide_i to signify different dimensions. As a result, the quantile prediction model regards $[e_{t-k}, \dots, e_{t-1}, ide_i]$ as input and predict the quantile of future errors.

The original algorithm requires to retrain quantile prediction in every step using the most recent data. However, it is time consuming to do that and we choose to retrain this model every 200 steps. And $[e_{t-k-j,i,j}, \dots, e_{t-1-j,i,j}, ide_i]$ is regarded as input because $e_{t-1,i,j}$ can not be observed until time step $t + j$. The lag k was 20.

ACI (Gibbs and Candes 2021)

This method represents the most classical online conformal prediction approach. Specifically, it adjusts the confidence interval based on whether the current interval covers the true value. Rather than directly modifying the interval length, it adaptively selects which quantile of the test set errors to use to construct the next interval. This process can be equivalently viewed as performing online gradient descent on the quantile parameter (Wu et al. 2025), where the loss function is the standard quantile loss.

SPCI (Xu and Xie 2023)

The core idea of this method involves using historical residuals to predict future residual quantiles, which closely resembles the approach of LPCI. However, the key distinction lies in the fact that SPCI was originally designed for univariate time series forecasting problems. In our experiments, we adapted this approach by calibrating separate future error quantile prediction models for each variable and each prediction horizon. All other experimental settings remained identical to those used in LPCI.

PID (Angelopoulos, Candes, and Tibshirani 2023)

This method applies control theory to the construction of confidence intervals for time series, consisting of three main components: The first part uses an algorithm similar to ACI to adaptively update the length of confidence intervals. Unlike traditional ACI which updates the interval length for the next time step based solely on whether the current time step is covered, this method employs PID control to incorporate coverage statistics from past periods when updating the interval length. And this procedure is the second part. The third part utilizes a model that predicts future errors based on past errors.

FCP (Teng et al. 2023)

The primary objective of this method is to shift traditional conformal prediction from computing residuals at the prediction level to the feature level. Specifically, for a deep learning model f , it can be decomposed into a feature extractor g and a prediction head h . Given an input-output pair (X, Y) , its latent feature representation is obtained through the feature extractor:

$$z = g(X) \quad (19)$$

Then, the optimal perturbed feature z' that minimizes the distance from the original feature while satisfying the prediction constraint can be calculated as follows:

$$\min_{z'} \|z' - z\| \quad \text{subject to} \quad h(z') = Y \quad (20)$$

This optimization is achieved through gradient descent in the feature space.

During calibration, the feature-space distances $\|z' - z\|$ for all validation samples are computed. Let Δ denote the $(1 - \alpha)$ -quantile of these distances across the validation set:

$$\Delta = \text{Quantile}_{1-\alpha}(\{\|z'_i - z_i\|\}_{i=1}^N) \quad (21)$$

For a test sample X_{N+1} with extracted feature $z_{N+1} = g(X_{N+1})$, we construct the prediction interval as:

$$\mathcal{S} = \{y \in \mathcal{Y} \mid \exists z' \text{ s.t. } y = h(z'), \|z' - z_{N+1}\| < \Delta\} \quad (22)$$

CF-SST (Chen, Ren, and Cheng 2024)

This method extends FCP. Since FCP was not designed for time series data, it only guarantees confidence interval coverage under exchangeable data assumptions. The CF-SST method extends the FCP to ensure coverage for time series data. Specifically, for the $(N + 1)$ -th data point: Traditional FCP uses the $1 - \alpha$ quantile of feature-space errors from the validation set, but CF-SST employs a weighted quantile approach that first calculates the similarity between the $(N + 1)$ -th sample's errors and those of each validation sample, then weights the validation set feature-space errors based on these similarity scores to compute a weighted quantile.

QCP (Romano, Patterson, and Candes 2019)

This method first converts the point prediction model into a quantile prediction model. It then evaluates the accuracy of this quantile prediction model by computing the errors between the predicted quantiles and the actual dependent variables on the validation set.

For test samples, the method constructs confidence intervals by combining two the quantile predictions from the quantile prediction model and the model’s observed errors on the validation set.

Discussion on baseline methods

All the baseline methods mentioned above are highly insightful and have made significant contributions to this field. Our method builds upon these existing approaches, and in the following discussion, we will focus on the methods most relevant to our work while highlighting the key differences between them and our proposed approach.

PID This method is insightful and somewhat similar to our approach. The paper about this method also mentions that using certain models to predict future error quantiles can reduce the variability of the adjust term, thereby improving prediction accuracy and efficiency. Our work advances beyond this in two key aspects. First, we demonstrate that features extracted by deep learning models can predict future error quantiles more effectively than relying solely on past error data. Second, through *MAE*-related theorems, we theoretically establish why using a model to predict future error quantiles yields superior confidence intervals. We further prove that better fitting of future error quantiles directly leads to improved confidence interval performance.

SPCI and LPCI These two methods also attempt to fit a predictive model for future error quantiles. And they rely on combining past errors to predict future error quantiles, which is similar to the PID approach. However, their coverage guarantees require this quantile prediction model to be accurate, thus imposing certain assumptions about the data distribution. Our experiments also show that these two methods often fail to achieve the target coverage rate. In the future, research on deep learning-based confidence intervals could consider both past errors and the features of the current sample to construct an error quantile prediction model. This would effectively combine the strengths of these existing methods and our proposed approach.

FCP This method also utilizes deep learning features to construct confidence intervals, but it was not specifically designed for time series data. Consequently, it cannot provide coverage guarantees for time series applications, a limitation that has been empirically confirmed in our experiments.

CF-SST This method modifies FCP to adapt it for time series data. However, our experiments reveal two key limitations: first, it still struggles to guarantee proper coverage rates, and second, the resulting confidence intervals tend to be unnecessarily wide. To theoretically explain these observations, let us first present the coverage guarantee theorem from the original paper. Suppose ΔCov represents the discrepancy between the achieved coverage rate and the target coverage level:

$$\Delta Cov \geq - \sum_{i=1}^n w^i d_{TV}(R(Z), R(Z)^i) \quad (23)$$

Where $R(Z)$ denotes the residual sequence from the first to the N -th data point $R(Z)^i$ represents the modified sequence obtained by swapping the residuals of the N -th and i -th data points in $R(Z)$, d_{TV} is the total variation norm between distributions. However, our task involves multi-step forecasting, where the output is a high-dimensional vector and consequently each residual is also high-dimensional. For high-dimensional random vectors, the total variation distance between them will be larger than the TV distance between their individual dimensions. This makes it difficult to control the TV distance between $R(z)$ and $R(z)^i$, ultimately leading to challenges in achieving the desired coverage rate.

Another challenge lies in CF-SST’s objective to find the optimal feature z' in the latent space such that when passed through the prediction head, it closely approximates the true output y . For one-dimensional prediction problems, it may be feasible to find such z' within a small neighborhood around the original feature z . However, for high-dimensional prediction tasks, finding z' becomes significantly more difficult – it is likely that z' may end up being very far from z , which would result in wide confidence intervals.

Certainly, there exist numerous other conformal prediction algorithms for time series, such as EnbPI (Xu and Xie 2021), NexCP (Barber et al. 2023), HopCPT (Auer et al. 2023), DtACI (Gibbs and Candès 2024), AgACI (Zaffran et al. 2022), MVP (Bastani et al. 2022), decay-OGD (Angelopoulos, Barber, and Bates 2024), SAOCP (Bhatnagar et al. 2023), Bellman conformal (Yang, Candès, and Lei 2024), etc. While our experiments do not cover all possible time series conformal prediction methods, we have included the most relevant baselines given our approach’s key components: Feature-based confidence interval prediction (represented by FCP and CF-SST), online optimization for interval width adjustment (represented by PID, ACI, and ECI), we believe our experimental comparisons are sufficiently comprehensive for evaluating our method’s effectiveness.

While all the aforementioned baseline methods represent groundbreaking contributions to the field of time series conformal prediction, our approach demonstrates particular suitability for the specific challenges posed by deep learning-based multivariate multi-step forecasting scenarios. The comparative advantages observed in our experiments should be interpreted as context-dependent refinements rather than universal superiority, as different methodological designs naturally excel in different application contexts. We further acknowledge the complementary value of advanced online optimization strategies from recent work, such as the expert aggregation approaches exemplified by DtACI (Gibbs and Candès 2024) and AgACI (Zaffran et al. 2022). These innovations offer promising directions for enhancing our framework.

Proofs

Lemma .5. Under assumption 0.1, for any t, i, j , we have:

$$-M - \gamma \leq a_{t,i,j} \leq M + \gamma \quad (24)$$

Proof. We use inductive method to proof that. First, as we defined in method, $a_{t,j} = 0$ when $t < j + 1$, so they conform 24. Then we prove that, if $-M - \gamma \leq a_{k,i,j} \leq M + \gamma$ holds for all $k \leq t$, then $-M - \gamma \leq a_{t+1,i,j} \leq M + \gamma$.

Case 1: If $-M - \gamma \leq a_{t-j,i,j} \leq -M$, then: $\hat{q}_{t-j,i,j} + a_{t-j,j} \leq 0$, and the prediction set is empty set. As a result:

$$a_{t+1,i,j} = a_{t,i,j} + \gamma(1 - \mathbb{I}_{y_{t-j,i,j} \in C_{t-j,i,j}} - \alpha) = a_{t,i,j} + \gamma(1 - \alpha) > a_{t-j,j} \geq -M - \gamma$$

And $a_{t+1,i,j}$ also less then $M + \gamma$.

Case 2: If $-M < a_{t-j,i,j} < M$, then:

$$a_{t+1,i,j} = a_{t,i,j} + \gamma(1 - \mathbb{I}_{y_{t-j,i,j} \in C_{t-j,i,j}} - \alpha) \leq a_{t,i,j} + \gamma(1 - \alpha) \leq M + \gamma(1 - \alpha)$$

$$a_{t+1,i,j} = a_{t,i,j} + \gamma(1 - \mathbb{I}_{y_{t-j,i,j} \in C_{t-j,i,j}} - \alpha) \geq a_{t,i,j} - \gamma\alpha \geq -M - \gamma$$

Case 3: If $M < a_{t-j,i,j} < M + \gamma$, then, because assumption 0.1, $\mathbb{I}_{y_{t-j,i,j} \in C_{t-j,i,j}}$ must be 1. As a result:

$$a_{t+1,i,j} = a_{t,i,j} - \gamma\alpha \leq M + \gamma$$

And $a_{t+1,i,j}$ also greater then $-\gamma$.

Combining the three cases, we can prove $a_{t+1,i,j}$ obeys (24), and complete the proof. $\square$

Proof of Theorem 0.2

Proof.

$$\left| \frac{\sum_{t=1}^T \mathbb{I}_{y_{t,i,j} \in C_{t,i,j}}}{T} - (1 - \alpha) \right| = \left| \frac{\sum_{t=T-j}^T \mathbb{I}_{y_{t,i,j} \in C_{t,i,j}}}{T} + \frac{\sum_{t=1}^{T-j-1} \mathbb{I}_{y_{t,i,j} \in C_{t,i,j}}}{T} - (1 - \alpha) \right| \quad (25)$$

$$\leq \left| \frac{\sum_{t=1}^{T-j-1} \mathbb{I}_{y_{t,i,j} \in C_{t,i,j}}}{T} - (1 - \alpha) \right| + \frac{j+1}{T} \quad (26)$$

According to (11),

$$\mathbb{I}_{y_{t,i,j} \in C_{t,i,j}} = \frac{a_{t+j+1,i,j} - a_{t+j,i,j}}{\gamma} + (1 - \alpha)$$

Therefore,

$$\frac{\sum_{t=1}^{T-j-1} \mathbb{I}_{y_{t,i,j} \in C_{t,i,j}}}{T} = \frac{a_{T,i,j} - a_{j+1,i,j}}{T\gamma} + \frac{(1 - \alpha)(T - j - 1)}{T} \quad (27)$$

substitute into (26).

$$\left| \frac{\sum_{t=1}^T \mathbb{I}_{y_{t,i,j} \in C_{t,i,j}}}{T} - (1 - \alpha) \right| \leq \left| \frac{2(M + \gamma)}{T\gamma} + \frac{(j+1)(1 - \alpha)}{T} \right| + \frac{j+1}{T} \quad (28)$$

$$\leq 2\left(\frac{M + \gamma}{T\gamma} + \frac{j+1}{T}\right) \quad (29)$$

The first inequality is because $a_{t,i,j}$ is bounded according to Equation 24. $\square$

Proof of Theorem 0.4

Proof. Similar to Gibbs and Candès (2024), we first define $\beta_{t,i,j}$ as:

$$\beta_{t,i,j} = \min : \{\beta : y_{t,i,j} \in [\hat{y}_{t,i,j} - \beta_{t,i,j} - \hat{q}_{t,i,j}, \hat{y}_{t,i,j} + \beta_{t,i,j} + \hat{q}_{t,i,j}]\}$$

and we define $a_{t,i,j}^* + \hat{q}_{t,i,j}$ as the true $1 - \alpha$ quantile of $y_{t,i,j} - \hat{y}_{t,i,j}$, therefore,

$$P(y_{t,i,j} \in [\hat{y}_{t,i,j} - a_{t,i,j}^* - \hat{q}_{t,i,j}, \hat{y}_{t,i,j} + a_{t,i,j}^* + \hat{q}_{t,i,j}]) = 1 - \alpha$$

Then $q_{t,i,j}^* = a_{t,i,j}^* + \hat{q}_{t,i,j}$. Besides, define pinball loss as:

$$l_\alpha(\beta_{t,i,j}, a_{t,i,j}) = \max(\alpha(a_{t,i,j} - \beta_{t,i,j}), (1 - \alpha)(\beta_{t,i,j} - a_{t,i,j}))$$

Therefore,

$$-\frac{d}{da_{t,i,j}} l_\alpha(\beta_{t,i,j}, a_{t,i,j}) = 1 - \alpha - \mathbb{I}_{y_{t-j,i,j} \in C_{t-j,i,j}}$$

and the iterate equation (11) can be expressed as:

$$a_{t+1,i,j} = a_{t-1,i,j} - \gamma \frac{d}{da_{t-j,i,j}} l_{\alpha}(\beta_{t-j,i,j}, a_{t-j,i,j})$$

The proof can be divided as three parts.

First we bound $\sum_t l_{\alpha}(\beta_{t,i,j}, a_{t,i,j}) - l_{\alpha}(\beta_{t,i,j}, a_{t,i,j}^*)$ using similar calculations in Theorem 10.1 of Hazan (2016). Then, we can use this bound to bound $\sum_t (a_{t,i,j} - a_{t,i,j}^*)^2$. Finally, the bound of $MACE$ can be obtained.

Step 1

To make symbols more simple, we omit dimension index i and use $l(a_{t,j}), l(a_{t,j}^*)$ to represent $l_{\alpha}(\beta_{t,i,j}, a_{t,i,j}), l_{\alpha}(\beta_{t,i,j}, a_{t,i,j}^*)$. Because pinball loss in a convex function, then, for every step t :

$$l(a_{t,j}) - l(a_{t,j}^*) \leq l'(a_{t,j}) \times (a_{t,j} - a_{t,j}^*) \quad (30)$$

And for all t such that $t > j$:

$$(a_{t+1,j} - a_{t,j}^*)^2 = (a_{t,j} - \gamma l'(a_{t-j,j}) - a_{t,j}^*)^2 \quad (31)$$

$$= (a_{t,j} - a_{t,j}^*)^2 - 2\gamma l'(a_{t-j,j})(a_{t,j} - a_{t,j}^*) + (\gamma l'(a_{t-j,j}))^2 \quad (32)$$

substitute into (30), we have the following equation for each $t > j$

$$2(l(a_{t,j}) - l(a_{t,j}^*)) \leq \frac{(a_{t,j} - a_{t,j}^*)^2 - (a_{t+1,j} - a_{t+1,j}^*)^2}{\gamma} + \gamma(l'(a_{t-j,j}))^2$$

Then:

$$\sum_{t=1}^T [l(a_{t,j}) - l(a_{t,j}^*)] = \sum_{t=1}^j l(a_{t,j}) - l(a_{t,j}^*) + \sum_{t=j+1}^T l(a_{t,j}) - l(a_{t,j}^*) \quad (33)$$

$$\leq M(j+1) + \sum_{t=j}^{T-1} l(a_{t,j}) - l(a_{t,j}^*) \quad (34)$$

$$\leq M(j+1) + \frac{1}{2} \sum_{t=j}^{T-1} \left[\frac{(a_{t,j} - a_{t,j}^*)^2 - (a_{t+1,j} - a_{t+1,j}^*)^2}{\gamma} + \gamma(l'(a_{t-j,j}))^2 \right] \quad (35)$$

$$\leq M(j+1) + \frac{1}{2} \sum_{t=j}^{T-1} \left[\frac{(a_{t,j} - a_{t,j}^*)^2 - (a_{t+1,j} - a_{t+1,j}^*)^2}{\gamma} \right] + \gamma(T-j-1) \quad (36)$$

$$= M(j+1) + \frac{a_{j,j}^2 - a_{T,j}^2 + \sum_{t=j}^{T-1} [2a_{t+1,j}(a_{t+1,j}^* - a_{t,j}^*)]}{2\gamma} \quad (37)$$

$$+ \frac{-2a_{j,j}a_{j,j}^* + 2a_{T,j}a_{T,j}^*}{2\gamma} + \gamma(T-j-1) \quad (38)$$

$$\leq M(j+1) + \frac{c \sum_{t=j}^{T-1} 2a_{t+1,j}(a_{t+1,j}^* - a_{t,j}^*)}{2\gamma} + \gamma(T-j-1) \quad (39)$$

$$\leq M(j+1) + \sqrt{c(T-j-1) \sum_{t=j}^{T-1} (a_{t+1,j}^* - a_{t,j}^*)} \quad (40)$$

$$= M(j+1) + \sqrt{c_2(T-j-1) \sum_{t=j}^{T-1} (q_{t+1,j}^* - \hat{q}_{t+1,j}) - (q_{t,j}^* - \hat{q}_{t,j})} \quad (41)$$

$$(42)$$

Inequality (35) is because of Assumption 0.1. Inequality (40) is because $a/\gamma + \gamma b \leq \sqrt{ab}$. c is a constant, but can be of different values in different equations. And:

$$\sum_{t=j}^{T-1} |(q_{t+1,j}^* - \hat{q}_{t+1,j}) - (q_{t,j}^* - \hat{q}_{t,j})| = \sum_{t=j}^{T-1} \sqrt{[(q_{t+1,j}^* - \hat{q}_{t+1,j}) - (q_{t,j}^* - \hat{q}_{t,j})]^2} \quad (43)$$

$$(44)$$

Then it equals to:

$$\sum_{t=j}^{T-1} \sqrt{[(q_{t+1,j}^* - \hat{q}_{t+1,j})^2 + (q_{t,j}^* - \hat{q}_{t,j})^2 - 2(q_{t+1,j}^* - \hat{q}_{t+1,j})(q_{t,j}^* - \hat{q}_{t,j})]} \quad (45)$$

Therefore:

$$\sum_{t=j}^{T-1} |(q_{t+1,j}^* - \hat{q}_{t+1,j}) - (q_{t,j}^* - \hat{q}_{t,j})| = c(T-j-1)\sigma(q_j^* - \hat{q}_j) \quad (46)$$

σ means standard deviation, defined in Theorem 0.4. Therefore, we have:

$$\sum_{t=1}^T [l(a_{t,j}) - l(a_{t,j}^*)] \leq M(j+1) + c(T-j-1)\sigma(q_j^* - \hat{q}_j) \quad (47)$$

Step 2

In this step we first want to show the following equation holds for all t, i, j .

$$E_\beta[l_\alpha(\beta_{t,i,j}, a_{t,i,j}) - l_\alpha(\beta_{t,i,j}, a_{t,i,j}^*)] \leq \frac{p_2}{2}(a_{t,i,j} - a_{t,i,j}^*)^2 \quad (48)$$

where p_2 is the minimum value of the density function of β . Again, we omit t, i, j for simplicity. We proof this inequality when $a \leq a^*$, and if $a > a^*$, the proof is the same.

$$E[l_\alpha(\beta, a) - l_\alpha(\beta, a^*)] = E\{\alpha(a - a^*)\mathbb{I}_{\beta < a} + [(1 - \alpha)(\beta - \alpha) - \alpha(a^* - \beta)]\mathbb{I}_{a < \beta < a^*} \quad (49)$$

$$+ (1 - \alpha)(a^* - a)\mathbb{I}_{a^* \leq \beta}\} \quad (50)$$

$$= E[\alpha(a - a^*)] + (\beta - a)I_{a < \beta < a^*} + (a^* - a)I_{a^* < \beta} \quad (51)$$

$$= E(\beta - a)\mathbb{I}_{a < \beta < a^*} \quad (52)$$

$$= \int_a^{a^*} (\beta - a)p(\beta)d\beta \geq \frac{p_1}{2}(a - a^*)^2 \quad (53)$$

(51) is because $\mathbb{I}_{\beta < a} = 1 - I_{a < \beta < a^*} - I_{\beta \geq a^*}$ and (52) is because $E\mathbb{I}_{\beta \geq a^*} = \alpha$. Insert this inequality into (47), we have:

$$\sum_{t=1}^T (a_{t,j} - a_{t,j}^*)^2 \leq \frac{p_1}{2}[M(j+1) + c(T-j-1)\sigma(q_j^* - \hat{q}_j)] \quad (54)$$

Step 3

$$\sum_{t=1}^T |a_{t,j} - a_{t,j}^*| \leq \sqrt{T \sum_{t=1}^T (a_{t,j} - a_{t,j}^*)^2} \quad (55)$$

$$\leq \sqrt{c[TM(j+1) + T^2\sigma(q_j^* - \hat{q}_j)]} \quad (56)$$

The first inequality is because of Cauchy-Schwarz inequality. Then:

$$\frac{1}{T} \sum_{t=1}^T |(P(y_t \in C_t) - (1 - \alpha))| \leq \frac{1}{T} \sum_{t=1}^T L|a_{t,j} - a_{t,j}^*| \leq c\sqrt{\sigma(q_j^* - \hat{q}_j) + \frac{M(j+1)}{T}} \quad (57)$$

The first inequality is because $P(y_t \in [\hat{y}_{t,j} - \hat{q}_{t,j} - a_{t,j}, \hat{y}_{t,j} + \hat{q}_{t,j} + a_{t,j}])$ is L -Lipschitz to $a_{t,j}$ and $P(y_t \in [\hat{y}_{t,j} - \hat{q}_{t,j} - a_{t,j}^*, \hat{y}_{t,j} + \hat{q}_{t,j} + a_{t,j}^*]) = 1 - \alpha$

□

Full Results

Full forecast results

We report the full results in Table 5, Table 6 and Table 7. The overall conclusion is similar to the one we reported in the main text of the paper.

Table 5: Full forecast results I

Dataset		weather				traffic				electricity				solar			
Method	Base model	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t
CP	itransformer	94.6%	1.510	87.3%	93.6%	87.9%	1.074	51.4%	87.0%	86.3%	0.970	52.5%	85.6%	87.6%	1.271	84.1%	85.8%
	leddam	94.1%	1.439	83.4%	93.5%	87.9%	1.082	56.0%	87.3%	85.7%	0.940	24.5%	84.6%	86.9%	1.212	83.3%	85.7%
	softs	93.8%	1.464	84.4%	92.6%	88.4%	1.129	59.6%	87.2%	86.5%	0.987	25.9%	85.8%	87.2%	1.263	83.5%	86.2%
	avg	94.2%	1.471	85.0%	93.2%	88.1%	1.095	55.7%	87.1%	86.2%	0.966	34.3%	85.3%	87.3%	1.248	83.6%	85.9%
TQA-E	itransformer	86.4%	1.145	82.2%	83.5%	87.4%	1.170	77.8%	86.9%	89.1%	1.180	84.3%	88.2%	89.4%	1.669	89.0%	88.9%
	leddam	86.0%	1.083	83.8%	83.3%	88.9%	1.116	77.1%	88.4%	89.0%	1.154	84.0%	88.2%	89.4%	1.674	88.9%	89.0%
	softs	84.9%	1.096	81.0%	81.7%	89.1%	1.151	78.2%	88.4%	89.0%	1.176	83.8%	88.0%	89.3%	1.710	88.9%	88.9%
	avg	85.8%	1.108	82.3%	82.8%	88.5%	1.146	77.7%	87.9%	89.0%	1.170	84.0%	88.2%	89.3%	1.684	88.9%	88.9%
LPCI	itransformer	85.3%	0.868	79.6%	84.4%	88.2%	1.024	71.2%	87.5%	90.3%	1.133	69.9%	88.4%	79.6%	0.980	74.3%	77.7%
	leddam	84.6%	0.939	52.6%	71.1%	88.0%	1.066	59.6%	86.9%	88.0%	0.997	54.1%	84.8%	84.7%	1.195	74.5%	80.3%
	softs	84.3%	0.947	50.4%	73.7%	89.0%	1.173	62.2%	88.0%	88.6%	1.053	53.2%	86.3%	82.7%	1.143	74.8%	75.7%
	avg	84.7%	0.918	60.8%	76.4%	88.4%	1.088	64.4%	87.5%	89.0%	1.061	59.1%	86.5%	82.3%	1.106	74.5%	77.9%
SPCI	itransformer	88.5%	1.196	82.2%	82.4%	83.2%	1.067	77.3%	84.5%	88.9%	1.099	72.9%	86.3%	82.7%	0.959	80.5%	75.4%
	leddam	87.3%	1.127	84.9%	77.5%	90.2%	1.076	85.7%	87.6%	90.3%	1.040	72.8%	81.7%	87.6%	1.192	78.4%	80.0%
	softs	88.3%	1.180	78.4%	75.0%	90.1%	1.148	78.2%	87.8%	90.2%	1.089	74.3%	85.2%	88.1%	1.139	80.5%	74.9%
	avg	88.0%	1.167	81.8%	78.3%	87.9%	1.097	80.4%	86.6%	89.8%	1.076	73.3%	84.4%	86.2%	1.096	79.8%	76.8%
ACI	itransformer	87.0%	1.193	82.3%	84.6%	89.1%	1.319	85.1%	88.9%	89.1%	1.146	86.0%	88.8%	89.9%	1.517	89.6%	89.6%
	leddam	86.6%	1.098	83.8%	84.6%	89.0%	1.305	84.6%	88.8%	89.0%	1.125	86.2%	88.8%	90.1%	1.460	89.7%	89.8%
	softs	86.2%	1.111	78.1%	81.8%	89.0%	1.354	84.6%	88.7%	89.0%	1.143	85.8%	88.8%	90.0%	1.496	89.7%	89.8%
	avg	86.6%	1.134	81.4%	83.6%	89.0%	1.326	84.8%	88.8%	89.1%	1.138	86.0%	88.8%	90.0%	1.491	89.7%	89.7%
PID	itransformer	89.9%	1.231	88.1%	89.7%	89.4%	1.180	81.4%	88.7%	89.6%	1.115	83.1%	89.5%	90.9%	1.632	89.9%	90.0%
	leddam	89.9%	1.154	88.1%	89.6%	89.3%	1.183	80.5%	88.5%	89.6%	1.074	83.1%	89.5%	91.5%	1.738	90.1%	90.0%
	softs	89.8%	1.205	87.9%	89.5%	89.1%	1.193	80.6%	88.3%	89.6%	1.091	83.1%	89.4%	91.0%	1.684	89.7%	89.9%
	avg	89.9%	1.197	88.0%	89.6%	89.3%	1.185	80.9%	88.5%	89.6%	1.093	83.1%	89.4%	91.2%	1.685	89.9%	90.0%
ECI	itransformer	89.9%	1.219	87.4%	89.5%	88.4%	1.053	78.5%	87.8%	88.4%	1.100	82.7%	88.2%	88.9%	1.384	87.0%	87.5%
	leddam	89.8%	1.133	87.3%	89.5%	88.2%	1.059	76.7%	87.8%	88.4%	1.097	83.0%	88.1%	88.1%	1.283	86.9%	87.2%
	softs	89.8%	1.088	87.1%	89.3%	88.5%	1.171	77.8%	87.8%	88.4%	1.085	82.9%	88.1%	88.3%	1.332	86.9%	87.4%
	avg	89.8%	1.147	87.3%	89.4%	88.4%	1.094	77.7%	87.8%	88.4%	1.094	82.9%	88.1%	88.4%	1.333	86.9%	87.4%
CQR	itransformer	89.9%	1.014	83.1%	88.3%	88.3%	1.141	59.3%	87.9%	87.1%	0.999	74.1%	88.3%	86.1%	0.880	83.1%	78.7%
	leddam	86.3%	0.885	75.6%	84.6%	87.7%	1.063	60.1%	87.5%	87.7%	0.963	73.9%	88.0%	88.9%	0.869	85.1%	81.0%
	softs	87.3%	0.944	81.7%	84.3%	87.9%	1.061	59.9%	87.9%	88.5%	1.120	74.6%	87.6%	86.5%	0.941	82.8%	82.2%
	avg	87.8%	0.947	80.1%	85.8%	88.0%	1.088	59.8%	87.8%	87.8%	1.027	74.2%	87.9%	87.2%	0.896	83.7%	80.6%
FCP	itransformer	95.3%	1.536	76.4%	90.8%	84.8%	1.086	56.4%	81.8%	85.7%	1.017	72.9%	77.1%	88.3%	1.391	85.3%	79.2%
	leddam	93.8%	1.487	71.7%	88.2%	85.1%	1.062	61.2%	82.1%	84.2%	0.982	75.1%	82.1%	85.3%	1.230	81.6%	82.7%
	softs	92.7%	1.489	80.0%	85.3%	87.3%	1.102	59.2%	84.2%	85.1%	1.092	78.2%	83.1%	86.3%	1.223	84.5%	86.2%
	avg	93.9%	1.504	76.0%	88.1%	85.7%	1.083	58.9%	82.7%	85.0%	1.030	75.4%	80.8%	86.7%	1.282	83.8%	82.7%
CF-SST	itransformer	91.0%	1.242	79.2%	88.2%	86.5%	1.089	68.4%	83.9%	87.8%	1.029	81.8%	80.8%	88.2%	1.377	85.9%	79.0%
	leddam	90.3%	1.241	86.1%	88.4%	86.2%	1.092	69.1%	83.1%	87.1%	1.063	82.1%	80.4%	87.2%	1.288	84.1%	82.2%
	softs	90.8%	1.215	83.8%	82.3%	87.3%	1.101	69.3%	85.1%	86.4%	1.090	78.2%	86.2%	88.8%	1.302	86.1%	85.9%
	avg	90.7%	1.233	83.0%	86.3%	86.7%	1.094	68.9%	84.0%	87.1%	1.061	80.7%	82.5%	88.1%	1.322	85.4%	82.4%
FFDCI	itransformer	89.4%	1.061	88.2%	89.1%	88.9%	1.067	81.5%	87.8%	89.4%	1.105	86.7%	89.0%	89.8%	1.054	89.4%	89.6%
	leddam	89.3%	1.040	87.4%	88.8%	88.7%	1.026	77.2%	87.6%	89.1%	1.007	84.6%	88.6%	89.9%	1.037	89.5%	89.6%
	softs	89.2%	1.086	87.2%	88.9%	88.3%	1.048	78.0%	87.4%	89.3%	1.051	84.3%	88.9%	89.8%	1.072	89.4%	89.5%
	avg	89.3%	1.062	87.6%	88.9%	88.6%	1.047	78.9%	87.6%	89.3%	1.054	85.2%	88.8%	89.8%	1.054	89.4%	89.6%
Improve		6.3%				3.8%				0.6%				20.3%			

Table 6: Full forecast results II

Dataset		ETTh1				ETTh2				ETTM1				ETTM2			
Method	Base model	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t
CP	itransformer	94.3%	2.527	87.2%	93.0%	90.9%	1.413	82.4%	89.8%	92.2%	1.862	87.0%	91.3%	87.8%	0.997	83.6%	86.6%
	leddam	94.5%	2.603	87.4%	92.4%	91.1%	1.354	84.4%	89.6%	92.9%	1.812	88.6%	91.7%	87.6%	0.985	83.2%	87.5%
	softs	94.2%	2.484	86.9%	92.2%	91.0%	1.334	84.7%	89.5%	93.4%	1.820	88.9%	92.0%	87.3%	0.973	82.7%	86.8%
	avg	94.3%	2.538	87.2%	92.6%	91.0%	1.367	83.8%	89.6%	92.8%	1.831	88.2%	91.6%	87.6%	0.985	83.2%	86.9%
TQA-E	itransformer	88.7%	1.843	86.4%	87.9%	87.3%	1.452	84.8%	85.7%	88.3%	1.943	85.1%	87.1%	87.9%	1.179	86.7%	85.9%
	leddam	90.9%	2.008	89.3%	89.8%	87.3%	1.338	85.0%	85.3%	88.3%	1.804	86.2%	87.3%	87.5%	1.148	86.0%	85.4%
	softs	88.5%	1.823	85.3%	87.7%	87.7%	1.340	85.5%	85.8%	88.1%	1.774	85.8%	87.0%	87.3%	1.155	85.8%	85.4%
	avg	89.4%	1.891	87.0%	88.5%	87.4%	1.377	85.1%	85.6%	88.2%	1.840	85.7%	87.1%	87.5%	1.161	86.2%	85.6%
LPCI	itransformer	84.1%	1.348	80.0%	81.8%	82.5%	0.985	75.7%	81.5%	87.8%	1.694	85.3%	87.2%	83.1%	0.896	80.4%	82.6%
	leddam	92.0%	2.109	80.8%	90.1%	88.1%	1.307	63.3%	84.8%	85.3%	1.434	72.1%	82.9%	83.9%	1.022	57.3%	80.1%
	softs	87.4%	1.596	77.4%	85.5%	84.8%	1.141	65.0%	80.8%	85.0%	1.396	70.7%	81.8%	83.4%	0.997	59.2%	79.3%
	avg	87.8%	1.684	79.4%	85.8%	85.1%	1.144	68.0%	82.4%	86.0%	1.508	76.0%	84.0%	83.5%	0.972	65.6%	80.7%
SPCI	itransformer	88.2%	1.928	85.9%	77.0%	86.2%	0.944	82.8%	86.5%	90.5%	1.820	81.0%	86.5%	87.8%	0.924	87.2%	85.4%
	leddam	90.2%	2.069	81.3%	85.0%	88.8%	1.350	85.3%	86.8%	89.8%	1.801	84.2%	81.4%	87.6%	1.018	88.0%	77.0%
	softs	91.1%	1.841	87.1%	89.9%	86.6%	1.188	85.0%	84.6%	88.7%	1.699	85.6%	82.7%	87.0%	0.991	88.4%	79.4%
	avg	89.8%	1.946	84.8%	84.0%	87.2%	1.161	84.4%	86.0%	89.7%	1.773	83.6%	83.5%	87.5%	0.978	87.9%	80.6%
ACI	itransformer	88.1%	1.801	84.2%	87.3%	87.1%	1.430	85.4%	86.5%	88.9%	1.972	85.2%	88.3%	88.5%	1.220	87.2%	87.7%
	leddam	88.4%	1.775	86.3%	88.0%	87.0%	1.312	85.7%	86.2%	89.1%	1.835	86.6%	88.7%	88.1%	1.190	86.7%	86.9%
	softs	88.1%	1.801	84.2%	87.3%	87.2%	1.293	85.6%	86.4%	88.8%	1.799	85.6%	88.4%	87.9%	1.190	86.0%	86.9%
	avg	88.2%	1.793	84.9%	87.5%	87.1%	1.345	85.6%	86.3%	88.9%	1.869	85.8%	88.5%	88.2%	1.200	86.6%	87.2%
PID	itransformer	90.3%	1.818	85.4%	89.1%	91.2%	1.643	82.7%	89.4%	89.6%	1.795	88.2%	89.1%	89.0%	1.243	89.5%	89.8%
	leddam	90.3%	1.770	85.2%	89.2%	90.4%	1.455	82.2%	89.3%	89.8%	1.706	88.6%	89.3%	90.0%	1.216	89.6%	89.7%
	softs	90.4%	1.789	85.5%	89.2%	91.5%	1.602	83.1%	89.3%	89.8%	1.618	88.5%	89.4%	90.1%	1.236	89.5%	89.8%
	avg	90.3%	1.792	85.3%	89.2%	91.0%	1.567	82.7%	89.3%	89.7%	1.706	88.4%	89.3%	89.7%	1.232	89.5%	89.8%
ECI	itransformer	92.2%	2.186	86.7%	90.5%	89.7%	1.327	84.6%	88.8%	90.4%	1.755	88.9%	90.1%	88.1%	1.148	86.8%	87.4%
	leddam	92.5%	2.147	86.8%	90.1%	89.6%	1.252	85.4%	88.6%	90.7%	1.687	89.4%	90.4%	88.1%	1.214	86.8%	87.5%
	softs	92.2%	2.043	86.5%	90.1%	89.6%	1.240	85.5%	88.5%	90.8%	1.645	89.5%	90.4%	88.1%	1.213	86.8%	87.5%
	avg	92.3%	2.125	86.7%	90.3%	89.6%	1.273	85.2%	88.6%	90.6%	1.696	89.3%	90.3%	88.1%	1.192	86.8%	87.5%
CQR	itransformer	95.9%	2.271	94.2%	93.4%	90.7%	1.346	87.5%	89.5%	93.2%	1.786	91.5%	90.7%	87.7%	1.048	86.0%	88.4%
	leddam	95.4%	2.177	92.4%	90.9%	92.1%	1.370	89.3%	89.9%	92.8%	1.692	89.4%	89.6%	87.2%	1.052	86.6%	89.8%
	softs	95.7%	2.166	92.8%	91.5%	91.9%	1.354	87.6%	89.9%	92.7%	1.606	91.5%	90.6%	87.8%	1.044	87.0%	89.0%
	avg	95.7%	2.204	93.2%	91.9%	91.6%	1.357	88.1%	89.8%	92.9%	1.695	90.8%	90.3%	87.5%	1.048	86.5%	89.1%
FCP	itransformer	94.1%	2.238	87.3%	90.7%	90.8%	1.499	89.5%	93.2%	88.2%	1.565	81.0%	83.1%	87.0%	1.007	82.1%	94.6%
	leddam	93.2%	2.086	87.3%	91.0%	90.6%	1.477	80.4%	85.4%	92.5%	1.896	86.1%	88.8%	87.9%	1.067	76.1%	83.3%
	softs	95.2%	2.475	90.5%	93.3%	86.3%	1.336	78.3%	80.5%	91.3%	1.829	86.7%	86.8%	85.9%	0.957	81.9%	80.5%
	avg	94.2%	2.266	88.4%	91.7%	89.2%	1.437	82.8%	86.4%	90.6%	1.763	84.6%	86.2%	86.9%	1.011	80.1%	86.2%
CF-SST	itransformer	92.4%	1.980	86.2%	89.7%	90.3%	1.411	87.3%	90.8%	90.7%	1.786	84.0%	86.4%	88.0%	1.071	76.2%	83.4%
	leddam	92.4%	1.971	86.1%	89.8%	91.6%	1.571	82.6%	86.9%	91.9%	1.750	85.8%	88.5%	87.0%	0.964	72.0%	80.0%
	softs	92.4%	1.980	86.2%	89.7%	86.3%	1.387	77.2%	81.1%	88.0%	1.736	84.7%	87.4%	87.4%	1.175	71.4%	84.4%
	avg	92.4%	1.977	86.2%	89.8%	89.4%	1.456	82.4%	86.3%	90.2%	1.757	84.8%	87.5%	87.4%	1.070	73.2%	82.6%
FFDCI	itransformer	89.9%	1.758	87.5%	88.3%	87.9%	1.336	85.7%	86.9%	89.6%	1.718	88.5%	89.3%	89.1%	1.122	88.0%	88.4%
	leddam	88.8%	1.666	87.7%	87.6%	86.7%	1.133	82.8%	85.4%	89.4%	1.588	88.2%	89.1%	88.8%	1.104	87.8%	88.3%
	softs	88.8%	1.624	87.3%	86.6%	87.1%	1.113	82.7%	85.9%	89.3%	1.565	87.9%	89.0%	88.7%	1.114	87.7%	88.1%
	avg	89.2%	1.683	87.5%	87.5%	87.3%	1.194	83.8%	86.0%	89.4%	1.623	88.2%	89.1%	88.9%	1.114	87.8%	88.3%
Improve				6.1%				-				4.2%				6.6%	

Table 7: Full forecast results III

Dataset		PEMS03				PEMS04				PEMS07				PEMS08			
Method	Base model	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t
CP	itransformer	84.4%	1.080	42.4%	82.3%	89.4%	1.296	72.3%	88.4%	90.2%	1.113	18.0%	89.6%	89.3%	1.163	84.0%	88.8%
	leddam	84.0%	0.992	39.8%	81.6%	86.7%	1.036	72.3%	84.8%	90.3%	1.065	27.2%	89.4%	88.2%	1.176	62.9%	86.6%
	softs	87.4%	1.312	57.3%	86.0%	90.1%	1.602	70.9%	89.3%	90.1%	1.320	37.5%	89.4%	89.3%	1.414	68.2%	88.6%
	avg	85.3%	1.128	46.5%	83.3%	88.7%	1.311	71.8%	87.5%	90.2%	1.166	27.6%	89.5%	88.9%	1.251	71.7%	88.0%
TQA-E	itransformer	88.7%	1.248	82.6%	88.1%	89.8%	1.302	81.2%	89.6%	89.3%	1.133	73.1%	88.5%	90.0%	1.242	79.4%	89.8%
	leddam	88.5%	1.167	77.8%	88.0%	89.2%	1.131	83.5%	88.7%	90.1%	1.053	77.6%	90.0%	89.4%	1.207	76.5%	89.0%
	softs	89.2%	1.403	84.7%	88.8%	90.0%	1.581	83.3%	89.8%	90.1%	1.322	83.3%	90.0%	89.7%	1.414	76.5%	89.5%
	avg	88.8%	1.272	81.7%	88.3%	89.7%	1.338	82.7%	89.4%	89.9%	1.169	78.0%	89.5%	89.7%	1.288	77.5%	89.4%
LPCI	itransformer	86.6%	1.201	55.6%	82.7%	87.5%	1.239	53.6%	84.4%	88.4%	1.074	42.7%	87.7%	89.3%	1.320	62.3%	84.7%
	leddam	87.3%	1.188	57.0%	81.7%	88.4%	1.156	60.8%	83.1%	89.4%	1.081	43.4%	85.0%	88.1%	1.182	50.1%	82.2%
	softs	86.9%	1.377	64.3%	79.2%	88.2%	1.591	54.5%	81.2%	89.6%	1.447	45.3%	83.8%	88.5%	1.519	59.4%	81.7%
	avg	86.9%	1.255	59.0%	81.2%	88.0%	1.329	56.3%	82.9%	89.1%	1.201	43.8%	85.5%	88.6%	1.341	57.3%	82.9%
SPCI	itransformer	87.6%	1.261	79.3%	81.5%	88.1%	1.282	80.4%	82.3%	88.2%	1.072	69.4%	91.2%	87.8%	1.328	86.2%	84.3%
	leddam	90.5%	1.221	80.2%	78.5%	90.9%	1.153	80.7%	82.5%	87.7%	1.054	80.8%	85.9%	89.4%	1.225	83.5%	86.6%
	softs	87.9%	1.256	86.4%	79.0%	89.9%	1.559	82.3%	78.6%	89.1%	1.444	71.3%	80.4%	88.5%	1.506	81.4%	80.0%
	avg	88.7%	1.246	82.0%	79.7%	89.6%	1.332	81.2%	81.2%	88.3%	1.190	73.8%	85.8%	88.6%	1.353	83.7%	83.6%
ACI	itransformer	87.9%	1.487	85.7%	87.4%	87.3%	1.477	81.1%	86.6%	88.8%	1.229	84.6%	88.4%	87.3%	1.477	81.1%	86.6%
	leddam	87.7%	1.413	84.1%	87.2%	87.5%	1.384	83.4%	87.2%	88.5%	1.177	78.0%	87.7%	87.5%	1.384	83.4%	87.2%
	softs	87.5%	1.637	83.8%	86.7%	87.1%	1.742	80.2%	86.5%	88.3%	1.450	78.5%	87.3%	87.1%	1.742	80.2%	86.5%
	avg	87.7%	1.512	84.5%	87.1%	87.3%	1.534	81.5%	86.8%	88.5%	1.285	80.4%	87.8%	87.3%	1.534	81.5%	86.8%
PID	itransformer	89.4%	1.303	85.2%	89.1%	89.8%	1.314	83.6%	89.5%	90.3%	1.144	81.2%	90.1%	90.1%	1.286	81.3%	89.3%
	leddam	89.4%	1.234	83.6%	88.8%	89.7%	1.178	84.9%	89.5%	90.1%	1.066	82.0%	89.7%	91.0%	1.362	81.5%	90.3%
	softs	89.4%	1.437	85.4%	89.0%	90.2%	1.616	85.4%	89.6%	90.5%	1.375	84.7%	90.1%	89.7%	1.449	79.4%	88.8%
	avg	89.4%	1.325	84.7%	89.0%	89.9%	1.369	84.6%	89.5%	90.3%	1.195	82.6%	90.0%	90.3%	1.366	80.7%	89.5%
ECI	itransformer	87.6%	1.199	81.8%	87.0%	88.8%	1.246	80.3%	88.6%	89.1%	1.084	73.3%	89.0%	89.0%	1.193	78.6%	88.7%
	leddam	87.5%	1.122	77.0%	86.9%	88.3%	1.138	82.7%	87.8%	89.1%	1.067	77.1%	88.9%	88.5%	1.163	75.9%	88.0%
	softs	88.0%	1.341	83.8%	87.5%	88.9%	1.513	82.4%	88.5%	88.9%	1.257	82.6%	88.7%	88.7%	1.354	75.7%	88.3%
	avg	87.7%	1.221	80.9%	87.2%	88.7%	1.299	81.8%	88.3%	89.1%	1.136	77.7%	88.9%	88.7%	1.236	76.8%	88.3%
CQR	itransformer	84.4%	1.056	64.3%	81.6%	89.0%	1.190	70.4%	88.3%	90.2%	1.104	39.5%	89.9%	88.7%	1.185	66.5%	87.9%
	leddam	86.1%	1.413	60.3%	84.1%	88.8%	1.310	76.0%	87.7%	90.4%	1.309	51.8%	89.9%	87.0%	1.344	68.7%	85.3%
	softs	86.7%	1.284	78.5%	85.2%	89.3%	1.965	79.9%	88.3%	91.0%	1.760	69.8%	90.0%	88.2%	1.549	76.4%	87.0%
	avg	85.7%	1.251	67.7%	83.6%	89.0%	1.488	75.4%	88.1%	90.5%	1.391	53.7%	89.9%	88.0%	1.360	70.5%	86.7%
FCP	itransformer	83.1%	1.166	41.5%	71.5%	83.8%	1.292	65.2%	74.6%	90.4%	1.229	56.1%	82.4%	84.0%	1.178	66.4%	89.7%
	leddam	84.9%	1.175	42.3%	82.7%	83.7%	1.486	67.2%	81.4%	90.3%	1.142	54.2%	87.2%	85.2%	1.262	67.5%	78.6%
	softs	84.1%	1.214	40.5%	74.3%	83.2%	1.523	63.2%	80.1%	90.1%	1.321	60.1%	86.2%	86.4%	1.128	66.2%	84.5%
	avg	84.0%	1.185	41.4%	76.2%	83.6%	1.434	65.2%	78.7%	90.3%	1.231	56.8%	85.3%	85.2%	1.190	66.7%	84.2%
CF-SST	itransformer	84.4%	1.168	70.2%	73.7%	84.7%	1.293	72.4%	76.7%	89.8%	1.234	75.5%	82.5%	86.6%	1.264	74.9%	80.8%
	leddam	85.1%	1.201	68.2%	72.5%	85.0%	1.502	72.7%	81.4%	90.1%	1.131	76.1%	87.3%	87.7%	1.364	71.1%	86.2%
	softs	85.0%	1.229	68.7%	77.3%	84.8%	1.602	74.3%	81.3%	90.0%	1.318	74.2%	88.0%	88.2%	1.241	75.2%	84.7%
	avg	84.8%	1.199	69.0%	74.5%	84.8%	1.466	73.1%	79.8%	90.0%	1.228	75.3%	85.9%	87.5%	1.290	73.7%	83.9%
FFDCI	itransformer	88.9%	1.139	83.5%	88.3%	89.4%	1.146	81.6%	88.9%	89.9%	1.048	80.1%	89.7%	89.1%	1.094	80.3%	88.6%
	leddam	88.4%	1.125	77.9%	87.7%	88.6%	1.083	83.3%	88.0%	89.6%	1.005	77.9%	89.3%	88.3%	1.100	76.2%	87.6%
	softs	88.9%	1.274	84.5%	88.0%	88.8%	1.359	81.0%	87.9%	89.5%	1.171	83.6%	89.1%	88.6%	1.235	76.3%	87.8%
	avg	88.7%	1.179	82.0%	88.0%	88.9%	1.196	82.0%	88.3%	89.6%	1.075	80.5%	89.4%	88.7%	1.143	77.6%	88.0%
Improve				7.3%				7.9%				5.4%				7.6%	

Full ablation experiment results

We report the full results of ablation experiments in Table 8.

We can conclude that not dynamically updating confidence interval typically results in a decline in coverage. For example, on the weather dataset, the coverage rate without updating confidence interval is less than 80%, whereas updating it increases the coverage to 89%. Moreover, updating the confidence interval can significantly improve coverage in the worst-performing dimension and the worst-performing prediction time step. For instance, in the electricity dataset, updating confidence interval raises the coverage in the worst-performing dimension from around 40% to 85%.

However, we must acknowledge that for certain datasets, such as the solar dataset, updating the confidence interval occasionally results in worse outcomes. This may be related to the inherent characteristics of the dataset, possibly because the solar dataset has relatively minor distribution shifts.

Our experimental results also demonstrate that disabling feature-based quantile fitting leads to degraded performance across all

Table 8: Forecast results of ablation experiments

Method		Ori				w/o update					w/o feature				
Dataset	Base model	<i>Cov</i>	<i>l</i>	<i>Min_d</i>	<i>Min_t</i>	<i>Cov</i>	<i>l</i>	<i>Min_d</i>	<i>Min_t</i>	<i>C_loss</i>	<i>Cov</i>	<i>l</i>	<i>Min_d</i>	<i>Min_t</i>	<i>l_loss</i>
weather	itransformer	89.4%	1.061	88.2%	89.1%	81.5%	0.842	70.6%	76.6%	7.9%	90.5%	1.170	89.2%	90.2%	10.3%
	leddam	89.3%	1.040	87.4%	88.8%	78.8%	0.776	64.6%	75.1%	10.5%	90.5%	1.095	89.1%	90.2%	5.2%
	SOFTS	89.2%	1.086	87.2%	88.9%	78.3%	0.795	61.4%	76.0%	10.9%	90.4%	1.144	89.0%	90.2%	5.4%
	ave	89.3%	1.062	87.6%	88.9%	79.5%	0.804	65.5%	75.9%	9.8%	90.5%	1.137	89.1%	90.2%	7.0%
traffic	itransformer	88.9%	1.067	81.5%	87.8%	85.7%	0.996	59.5%	80.4%	3.2%	89.4%	1.137	83.1%	88.9%	6.5%
	leddam	88.7%	1.026	77.2%	87.6%	85.6%	0.953	59.0%	80.7%	3.1%	89.2%	1.141	82.4%	88.9%	11.2%
	SOFTS	88.3%	1.048	78.0%	87.4%	82.4%	0.928	49.9%	78.5%	5.9%	89.4%	1.172	82.5%	88.8%	11.8%
	ave	88.6%	1.047	78.9%	87.6%	84.6%	0.959	56.1%	79.9%	4.0%	89.3%	1.150	82.6%	88.9%	9.8%
electricity	itransformer	89.4%	1.105	86.7%	89.0%	89.2%	1.102	69.0%	86.7%	0.2%	89.6%	1.151	86.8%	89.5%	4.2%
	leddam	89.1%	1.007	84.6%	88.6%	84.5%	0.880	25.3%	81.8%	4.6%	89.6%	1.066	86.3%	89.4%	5.9%
	SOFTS	89.3%	1.051	84.3%	88.9%	87.0%	0.981	24.9%	85.5%	2.3%	89.6%	1.087	87.1%	89.4%	3.4%
	ave	89.3%	1.054	85.2%	88.8%	86.9%	0.988	39.8%	84.7%	2.4%	89.6%	1.102	86.7%	89.4%	4.5%
solar	itransformer	89.8%	1.054	89.4%	89.6%	88.9%	1.028	86.6%	85.3%	0.9%	89.7%	1.490	89.2%	89.6%	41.4%
	leddam	89.9%	1.037	89.5%	89.6%	90.5%	1.027	86.0%	82.2%	-0.6%	89.6%	1.476	89.2%	89.5%	42.3%
	SOFTS	89.8%	1.072	89.4%	89.5%	86.6%	1.041	82.2%	77.6%	3.2%	89.6%	1.519	89.1%	89.5%	41.7%
	ave	89.8%	1.054	89.4%	89.6%	88.7%	1.032	84.9%	81.7%	1.1%	89.7%	1.495	89.2%	89.5%	41.8%
ETTm1	itransformer	89.6%	1.718	88.5%	89.3%	84.3%	1.473	79.4%	82.1%	5.3%	90.2%	1.770	89.2%	90.0%	3.1%
	leddam	89.4%	1.588	88.2%	89.1%	81.4%	1.259	74.7%	78.4%	8.0%	90.3%	1.668	89.5%	90.2%	5.1%
	SOFTS	89.3%	1.565	87.9%	89.0%	79.6%	1.198	70.8%	77.1%	9.7%	90.4%	1.635	89.5%	90.2%	4.5%
	ave	89.4%	1.623	88.2%	89.1%	81.8%	1.310	75.0%	79.2%	7.7%	90.3%	1.691	89.4%	90.1%	4.2%
ETTm2	itransformer	89.1%	1.122	88.0%	88.4%	84.4%	0.888	77.9%	77.7%	4.7%	89.7%	1.170	89.2%	89.4%	4.3%
	leddam	88.8%	1.104	87.8%	88.3%	80.4%	0.771	75.5%	77.6%	8.4%	89.7%	1.153	89.3%	89.4%	4.4%
	SOFTS	88.7%	1.114	87.7%	88.1%	79.6%	0.766	74.7%	76.8%	9.1%	89.7%	1.156	89.2%	89.4%	3.8%
	ave	88.9%	1.114	87.8%	88.3%	81.5%	0.808	76.0%	77.4%	7.4%	89.7%	1.160	89.2%	89.4%	4.1%
ETTth1	itransformer	89.9%	1.758	87.5%	88.3%	88.9%	1.723	85.0%	86.4%	1.0%	93.0%	2.224	88.4%	90.4%	26.5%
	leddam	88.6%	1.666	87.7%	87.6%	87.3%	1.606	84.7%	85.3%	1.5%	93.3%	2.288	88.5%	90.2%	37.3%
	SOFTS	88.8%	1.624	87.3%	86.6%	87.0%	1.561	85.2%	80.9%	1.9%	93.0%	2.189	88.3%	90.2%	34.8%
	ave	89.2%	1.683	87.5%	87.5%	87.7%	1.630	85.0%	84.2%	1.4%	93.1%	2.234	88.4%	90.3%	32.7%
ETTth2	itransformer	87.9%	1.336	85.7%	86.9%	81.1%	1.072	59.6%	71.1%	6.8%	89.7%	1.354	87.0%	89.1%	1.3%
	leddam	86.7%	1.133	82.8%	85.4%	84.4%	1.037	74.9%	81.3%	2.4%	89.5%	1.261	86.8%	89.0%	11.2%
	SOFTS	87.1%	1.113	82.7%	85.9%	84.2%	1.011	74.0%	81.1%	2.9%	89.6%	1.257	87.1%	88.8%	12.9%
	ave	87.3%	1.194	83.8%	86.0%	83.2%	1.040	69.5%	77.8%	4.0%	89.6%	1.290	86.9%	89.0%	8.1%
PEMS03	itransformer	88.9%	1.139	83.5%	88.3%	81.6%	0.963	44.5%	76.5%	7.2%	89.4%	1.304	86.2%	89.2%	14.5%
	leddam	88.4%	1.125	77.9%	87.7%	80.3%	0.902	38.4%	76.3%	8.1%	89.2%	1.222	84.4%	88.9%	8.6%
	SOFTS	88.9%	1.274	84.5%	88.0%	82.9%	1.101	70.9%	77.0%	6.0%	89.5%	1.444	86.7%	89.4%	13.3%
	ave	88.7%	1.179	82.0%	88.0%	81.6%	0.989	51.3%	76.6%	7.1%	89.4%	1.323	85.8%	89.2%	12.2%
PEMS04	itransformer	89.4%	1.146	81.6%	88.9%	86.4%	1.076	49.8%	83.0%	2.9%	89.8%	1.311	85.2%	89.7%	14.5%
	leddam	88.6%	1.083	83.3%	88.0%	83.0%	0.930	68.5%	80.0%	5.6%	89.6%	1.175	85.3%	89.5%	8.5%
	SOFTS	88.8%	1.359	81.0%	87.9%	85.2%	1.247	69.2%	82.1%	3.6%	89.8%	1.582	85.2%	89.7%	16.4%
	ave	88.9%	1.196	82.0%	88.3%	84.9%	1.085	62.5%	81.7%	4.0%	89.7%	1.356	85.3%	89.6%	13.4%
PEMS07	itransformer	89.9%	1.048	80.1%	89.7%	89.6%	1.038	37.0%	88.1%	0.4%	90.1%	1.123	83.4%	90.0%	7.2%
	leddam	89.6%	1.005	77.9%	89.3%	87.1%	0.927	31.2%	85.2%	2.5%	90.1%	1.071	83.9%	90.0%	6.6%
	SOFTS	89.5%	1.171	83.6%	89.1%	86.6%	1.079	44.3%	84.8%	2.8%	90.1%	1.339	85.3%	90.0%	14.3%
	ave	89.6%	1.075	80.5%	89.4%	87.8%	1.014	37.5%	86.0%	1.9%	90.1%	1.178	84.2%	90.0%	9.6%
PEMS08	itransformer	89.1%	1.094	80.3%	88.6%	85.4%	1.015	61.1%	82.4%	3.7%	89.9%	1.250	83.2%	89.8%	14.3%
	leddam	88.3%	1.100	76.2%	87.6%	82.7%	0.953	54.9%	79.7%	5.6%	89.8%	1.232	81.7%	89.6%	12.0%
	SOFTS	88.6%	1.235	76.3%	87.8%	83.6%	1.106	63.6%	79.3%	5.0%	89.8%	1.428	81.1%	89.6%	15.6%
	ave	88.7%	1.143	77.6%	88.0%	83.9%	1.025	59.9%	80.5%	4.7%	89.8%	1.303	82.0%	89.7%	14.0%

datasets. While the coverage rates remain relatively stable, we observe 4-40% increases in confidence interval lengths. Notably, the magnitude of this degradation directly correlates with our method’s performance advantage over baselines.

A representative case is the solar dataset, where our method achieves its most significant improvement over baselines – here, feature-based error fitting reduces interval widths by over 40%. The degree of improvement varies across datasets, suggesting this effect is closely tied to dataset-specific characteristics.

Sensitive analysis

Sensitive analysis of hidden size

Besides, we replaced the hidden size in quantile prediction model by 256,1024 and conducted additional experiments, resulting in Table 9. We also color the best performance **red**.

Table 9: Results of sensitive analysis experiments of hidden size

Hidden size		256				512				1024			
Dataset	Base model	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t
ETTh1	iTransformer	89.3%	1.681	86.6%	87.9%	89.9%	1.758	87.5%	88.3%	89.8%	1.703	87.4%	88.4%
	Leddiam	89.3%	1.684	87.4%	87.2%	88.8%	1.666	87.7%	87.6%	88.5%	1.648	87.3%	85.7%
	SOFTS	89.7%	1.691	87.6%	88.2%	88.8%	1.624	87.3%	86.6%	89.2%	1.653	87.3%	87.6%
ETTh2	iTransformer	87.7%	1.357	84.3%	85.4%	87.9%	1.336	85.7%	86.9%	86.5%	1.241	82.6%	84.4%
	Leddiam	86.9%	1.145	83.0%	85.7%	86.7%	1.133	82.8%	85.4%	86.2%	1.117	82.4%	84.9%
	SOFTS	87.3%	1.120	83.0%	86.2%	87.1%	1.113	82.7%	85.9%	86.9%	1.100	82.5%	85.5%
ETTm1	iTransformer	89.6%	1.698	88.7%	89.2%	89.6%	1.718	88.5%	89.3%	89.5%	1.690	88.3%	89.1%
	Leddiam	89.5%	1.613	88.3%	89.0%	89.4%	1.588	88.2%	89.1%	89.2%	1.582	88.0%	88.9%
	SOFTS	89.5%	1.591	88.2%	89.1%	89.3%	1.565	87.9%	89.0%	89.3%	1.555	87.9%	88.8%
ETTm2	iTransformer	89.2%	1.163	88.3%	88.6%	89.1%	1.122	88.0%	88.4%	89.3%	1.155	88.1%	88.5%
	Leddiam	88.9%	1.116	87.9%	88.3%	88.8%	1.104	87.8%	88.3%	88.7%	1.104	87.7%	88.1%
	SOFTS	88.8%	1.121	87.8%	88.3%	88.7%	1.114	87.7%	88.1%	88.7%	1.115	87.7%	88.1%
PEMS08	iTransformer	89.1%	1.126	80.5%	88.6%	89.1%	1.094	80.3%	88.6%	88.8%	1.097	79.7%	88.3%
	Leddiam	88.4%	1.112	76.4%	87.5%	88.3%	1.100	76.2%	87.6%	88.1%	1.092	76.1%	87.1%
	SOFTS	88.7%	1.243	75.5%	87.8%	88.6%	1.235	76.3%	87.8%	88.7%	1.239	75.6%	88.0%
solar	iTransformer	89.9%	1.084	89.6%	89.6%	89.8%	1.054	89.4%	89.6%	89.7%	1.040	89.4%	89.4%
	Leddiam	89.8%	0.984	89.5%	89.5%	89.9%	1.037	89.5%	89.6%	89.8%	1.005	89.5%	89.7%
	SOFTS	89.7%	0.967	89.2%	89.3%	89.8%	1.072	89.4%	89.5%	89.8%	1.074	89.3%	89.5%
weather	iTransformer	89.4%	1.075	88.3%	89.0%	89.4%	1.061	88.2%	89.1%	89.5%	1.096	88.3%	89.0%
	Leddiam	89.3%	1.067	87.6%	88.9%	89.3%	1.040	87.4%	88.8%	89.3%	1.046	87.2%	88.9%
	SOFTS	89.3%	1.093	87.3%	88.8%	89.2%	1.086	87.2%	88.9%	89.3%	1.110	87.2%	88.8%

It can be concluded from this table that the performance of our method is generally stable with different hidden sizes. And in many cases, increasing the hidden size to 1024 might slightly improve the results.

Results of other methods

We also conducted experiments about Monte-Carlo Dropout (Gal and Ghahramani 2015) because it is also able to provide confidence interval without changing the original model. In these experiments, the dropout rate is fixed as 0.3. And we use the point prediction models to predict 100 times, regarding the 5% and 95% quantiles of these predictions as the lower and upper bounds of confidence intervals. The results are shown in Table 10.

It is obvious that the coverages of Monte-Carlo Dropout are far away from 90%. This pattern is consistent with experiments in Lin, Trivedi, and Sun (2022); Xu and Xie (2021), where the coverages are also very low.

Although our work primarily focuses on how to obtain confidence intervals without altering the time series forecasting model, we still conducted additional experiments where we modified the point prediction model to a interval prediction model. Specifically, we added two linear layers to predict the 5% and 95% quantiles of the prediction, and then used quantile loss for training. The dataset division, optimizer, hyperparameters, and other settings were consistent with those used in the previous

Table 10: Results of Monte-Carlo Dropout

	iTransformer				Leddiam				SOFTS			
dataset	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t
weather	19.9%	0.144	4.9%	12.9%	27.1%	0.132	10.8%	23.5%	26.3%	0.117	13.0%	21.8%
electricity	28.7%	0.186	14.3%	25.9%	40.1%	0.257	13.8%	37.4%	27.9%	0.170	13.0%	24.2%
solar	40.4%	0.200	36.2%	35.7%	53.3%	0.221	45.1%	46.7%	55.3%	0.182	53.8%	48.9%
traffic	48.1%	0.245	7.5%	44.8%	59.7%	0.347	23.9%	56.8%	52.5%	0.250	23.1%	49.5%
ETTh1	28.3%	0.290	19.8%	26.3%	25.9%	0.265	17.0%	23.8%	24.5%	0.244	12.5%	22.3%
ETTh2	19.0%	0.172	10.3%	15.2%	23.1%	0.169	18.8%	21.0%	25.6%	0.180	20.2%	22.1%
ETTM1	33.5%	0.359	18.2%	30.9%	30.7%	0.315	19.9%	28.3%	31.0%	0.329	18.9%	28.9%
ETTM2	25.3%	0.185	17.4%	23.4%	27.5%	0.169	23.1%	25.2%	29.8%	0.167	23.3%	27.1%
PEMS08	34.2%	0.254	4.9%	31.0%	39.5%	0.326	12.0%	38.0%	37.3%	0.332	19.0%	35.8%
PEMS03	28.2%	0.244	11.2%	24.2%	34.9%	0.298	20.6%	32.3%	31.9%	0.303	21.6%	29.0%
PEMS04	27.4%	0.220	4.4%	24.3%	42.3%	0.336	15.5%	40.5%	33.5%	0.339	18.3%	31.3%
PEMS07	38.4%	0.252	0.4%	36.9%	46.4%	0.334	16.8%	44.5%	39.9%	0.327	15.8%	38.3%

experiments. After training the model, we deployed it on the test set and observed the prediction results, which are presented in Table 11. And the coverage below 50% is colored in purple.

Table 11: Results of quantile regression

	iTransformer				Leddiam				SOFTS			
dataset	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t
weather	76.3%	0.633	61.7%	72.0%	75.9%	0.638	58.6%	71.6%	78.3%	0.715	68.3%	73.4%
electricity	87.2%	0.935	76.5%	85.9%	85.5%	0.880	68.3%	83.2%	88.6%	0.971	73.1%	87.6%
solar	90.4%	0.897	87.3%	77.3%	87.4%	0.779	84.9%	75.3%	88.5%	0.860	86.3%	72.2%
traffic	87.2%	0.936	58.9%	85.6%	86.8%	0.977	66.1%	85.1%	88.9%	1.014	76.2%	86.8%
ETTh1	86.9%	1.485	82.6%	84.6%	89.5%	1.540	86.0%	88.3%	89.2%	1.529	84.9%	87.8%
ETTh2	85.8%	0.995	78.7%	82.7%	85.9%	1.083	82.0%	84.1%	86.2%	1.084	81.5%	84.1%
ETTM1	77.8%	1.035	74.9%	73.9%	82.8%	1.223	79.3%	80.7%	81.4%	1.187	71.2%	78.8%
ETTM2	83.4%	0.822	77.6%	80.1%	82.7%	0.793	78.8%	80.4%	85.7%	0.938	81.6%	84.3%
PEMS08	83.5%	0.994	23.2%	80.9%	78.1%	0.958	50.4%	74.5%	86.2%	1.266	66.1%	84.3%
PEMS03	78.3%	0.920	51.6%	73.5%	76.4%	0.827	47.1%	70.1%	81.9%	1.164	67.0%	78.4%
PEMS04	85.4%	1.063	37.9%	83.7%	84.4%	0.967	49.5%	81.8%	87.9%	1.453	73.7%	86.8%
PEMS07	87.5%	1.016	6.5%	86.3%	83.0%	0.796	29.1%	80.6%	88.2%	1.228	66.6%	87.0%

We can observe that directly using quantile regression makes it difficult to guarantee coverage. For example, the coverage for the weather dataset does not even reach 80%. Furthermore, when examining the coverage for the worst dimension and the worst prediction time step, the results are even poorer. For instance, in the case of the PEMS08 dataset, the coverage in the worst dimension can even fall below 50%. Similarly, for the solar dataset using SOFTS, although the overall coverage reaches 88.5%, the coverage for the worst time step drops to just 72%. These results suggest that time series datasets often suffer from distribution shift, meaning that the model trained on the training set is often not able to provide valid confidence intervals on the test set. Moreover, the degree of distribution shift may vary across different dimensions of the dataset. It can lead to a scenario where, although the overall coverage might still be relatively good, the coverage for the worst dimension is very low.

More Discussion on *MACE*

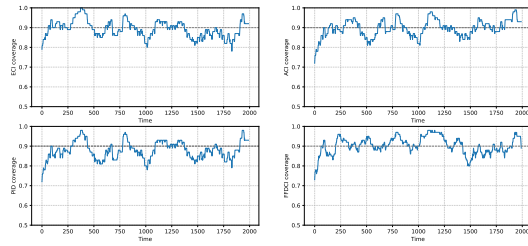
Furthermore, Figure 3 presents the temporal evolution of local coverage rates across different datasets for all evaluated algorithms. In each dataset-specific subplot of Figure 3, the smaller subplot in the bottom right corner represents the performance of our method, while the other three subplots display the baseline methods' performance. The figure also shows that our method maintains local coverage rates consistently closer to 90%.

Extended version of FFDCI

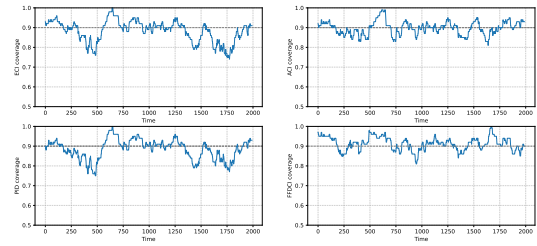
We have developed several extensions to our proposed algorithm: (1) The first extension adopts the ECI approach for updating confidence interval lengths (see Equation 16); (2) The second extension replaces the MLP with an LSTM model (with a single 512-dimensional hidden layer) for error quantile prediction, better suited for time series forecasting; (3) The third extension incorporates the SFOGD methodology, adaptively determining the update rate for confidence interval lengths using accumulated second-order gradient information. Specifically, we modify the parameter $a_{t+1,i,j}$ update rule as follows:

$$a_{t+1,i,j} = a_{t,i,j} + \frac{\gamma}{\sqrt{\sum_{i=1}^{t-j} (1 - \mathbb{I}_{y_{t-j,i,j} \in C_{t-j,i,j}} - \alpha)^2}} (1 - \mathbb{I}_{y_{t-j,i,j} \in C_{t-j,i,j}} - \alpha) \quad (58)$$

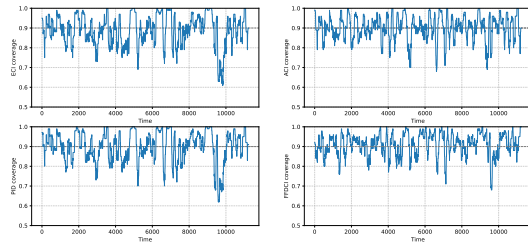
The following Table 12 presents the performance of the original algorithm and its three extended versions. We highlight the confidence interval lengths where extended versions outperform the original version in **red**, and **bold** the interval lengths of the best-performing method among all four approaches (three extended versions plus the original). Additionally, we report the frequency of each of these four methods achieving optimal results in the last row.



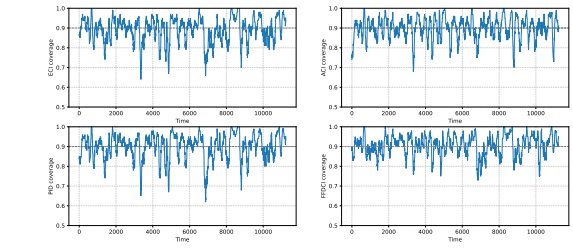
(a) ETTh1



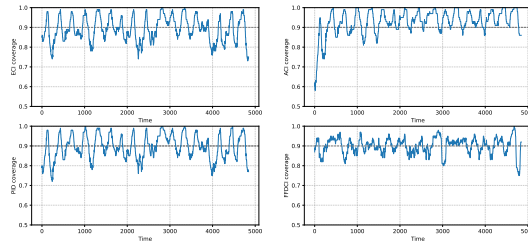
(b) ETTh2



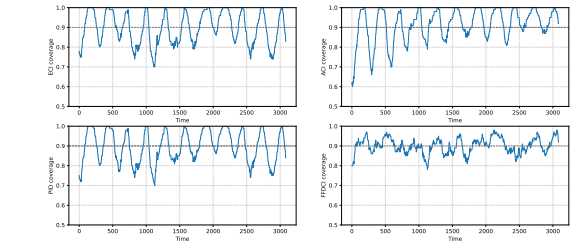
(c) ETTm1



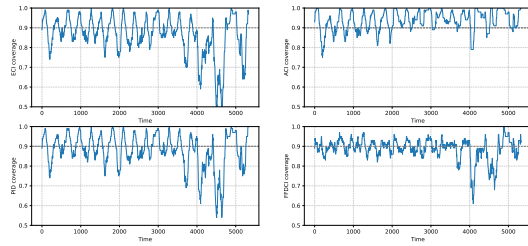
(d) ETTm2



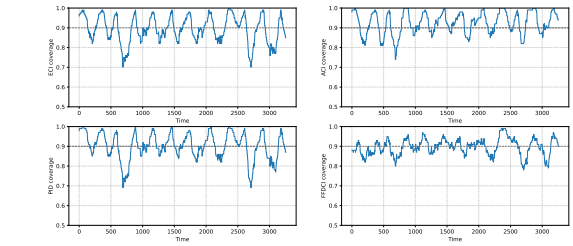
(e) PEMS03



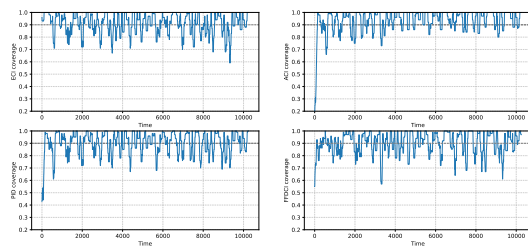
(f) PEMS04



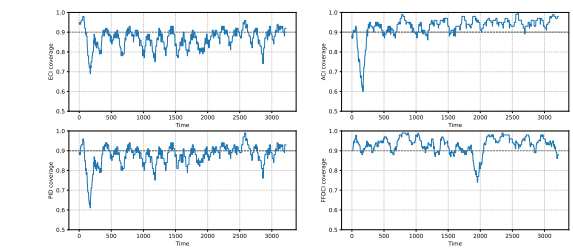
(g) PEMS07



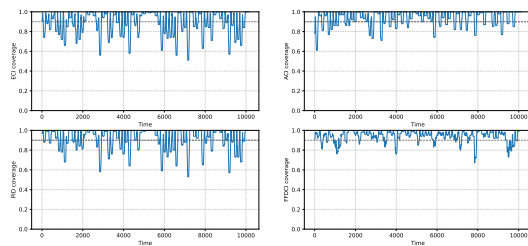
(h) PEMS08



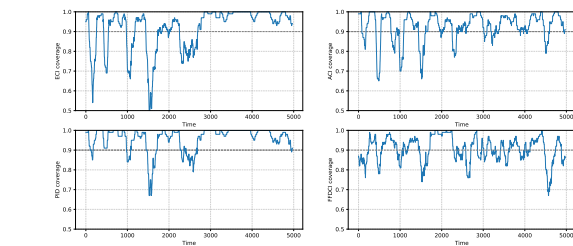
(i) weather



(j) traffic



(k) solar



(l) electricity

Table 12: Forecast results of extended FFDCI

Method		FFDCI-ori				FFDCI-ECI				FFDCI				FFDCI-SFOGD			
Dataset	Base model	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t	Cov	l	Min_d	Min_t
weather	itransformer	89.4%	1.061	88.2%	89.1%	89.5%	1.046	88.0%	88.7%	89.2%	1.119	87.3%	88.4%	91.2%	1.032	84.8%	88.4%
	leddam	89.3%	1.040	87.4%	88.8%	89.5%	1.074	87.2%	88.8%	89.3%	1.054	87.4%	88.9%	90.1%	1.051	81.6%	87.0%
	SOFTS	89.2%	1.086	87.2%	88.9%	89.4%	1.101	86.6%	89.0%	89.4%	1.116	87.3%	89.0%	90.0%	1.069	81.5%	87.6%
	ave	89.3%	1.062	87.6%	88.9%	89.5%	1.074	87.3%	88.8%	89.3%	1.096	87.3%	88.8%	90.4%	1.050	82.7%	87.7%
traffic	itransformer	88.9%	1.067	81.5%	87.8%	89.6%	1.076	82.2%	88.6%	88.2%	1.016	80.1%	87.7%	90.8%	1.105	70.9%	88.5%
	leddam	88.7%	1.026	77.2%	87.6%	89.2%	1.042	77.0%	88.3%	88.7%	1.028	77.5%	88.0%	90.1%	1.060	67.1%	88.3%
	SOFTS	88.3%	1.048	78.0%	87.4%	89.7%	1.085	78.5%	88.5%	88.4%	1.050	77.6%	87.7%	91.0%	1.119	69.1%	88.7%
	ave	88.6%	1.047	78.9%	87.6%	89.5%	1.068	79.3%	88.5%	88.4%	1.032	78.4%	87.8%	90.6%	1.095	69.1%	88.5%
electricity	itransformer	89.4%	1.105	86.7%	89.0%	91.0%	1.161	86.4%	90.4%	89.1%	1.100	85.1%	88.8%	93.7%	1.271	83.0%	92.5%
	leddam	89.1%	1.007	84.6%	88.6%	89.9%	1.042	83.2%	89.0%	88.7%	1.014	84.0%	88.4%	91.0%	1.073	65.6%	88.6%
	SOFTS	89.3%	1.051	84.3%	88.9%	89.7%	1.042	82.9%	89.2%	88.8%	1.024	83.9%	88.5%	90.8%	1.069	65.2%	89.3%
	ave	89.3%	1.054	85.2%	88.8%	90.2%	1.081	84.2%	89.5%	88.9%	1.046	84.3%	88.5%	91.8%	1.137	71.3%	90.1%
solar	itransformer	89.8%	1.054	89.4%	89.6%	91.1%	1.121	90.7%	90.5%	89.6%	1.036	89.2%	89.4%	96.2%	1.258	94.8%	94.8%
	leddam	89.9%	1.037	89.5%	89.6%	90.7%	0.979	89.8%	90.2%	89.8%	1.042	89.4%	89.6%	94.8%	1.109	92.5%	93.0%
	SOFTS	89.8%	1.072	89.4%	89.5%	90.5%	0.950	89.6%	90.0%	89.4%	0.963	88.9%	89.1%	95.0%	1.078	93.0%	93.9%
	ave	89.8%	1.054	89.4%	89.6%	90.8%	1.017	90.0%	90.2%	89.6%	1.013	89.1%	89.4%	95.3%	1.148	93.5%	93.9%
ETTm1	itransformer	89.6%	1.718	88.5%	89.3%	90.2%	1.740	88.9%	89.3%	89.3%	1.668	88.2%	89.0%	89.6%	1.688	85.9%	87.6%
	leddam	89.4%	1.588	88.2%	89.1%	89.6%	1.605	87.8%	88.7%	89.6%	1.622	88.3%	89.4%	88.3%	1.523	83.1%	85.2%
	SOFTS	89.3%	1.565	87.9%	89.0%	89.7%	1.585	87.2%	88.9%	89.4%	1.550	88.1%	89.2%	88.3%	1.508	81.4%	86.4%
	ave	89.4%	1.623	88.2%	89.1%	89.8%	1.643	88.0%	89.0%	89.4%	1.613	88.2%	89.2%	88.7%	1.573	83.5%	86.4%
ETTm2	itransformer	89.1%	1.122	88.0%	88.4%	89.6%	1.144	88.3%	88.8%	89.1%	1.133	88.2%	88.7%	91.1%	1.131	85.7%	88.7%
	leddam	88.8%	1.104	87.8%	88.3%	88.8%	1.087	87.5%	88.0%	88.8%	1.113	87.9%	88.3%	88.6%	1.003	83.5%	86.2%
	SOFTS	88.7%	1.114	87.7%	88.1%	88.7%	1.096	87.5%	87.9%	88.7%	1.110	87.7%	88.2%	88.5%	1.011	83.8%	86.0%
	ave	88.9%	1.114	87.8%	88.3%	89.0%	1.109	87.8%	88.2%	88.9%	1.119	87.9%	88.4%	89.4%	1.048	84.3%	87.0%
ETTth1	itransformer	89.9%	1.758	87.5%	88.3%	89.8%	1.758	87.0%	87.5%	90.4%	1.763	87.5%	89.6%	89.6%	1.747	85.9%	86.6%
	leddam	88.8%	1.666	87.7%	87.6%	89.8%	1.702	88.8%	87.9%	90.4%	1.892	87.8%	88.8%	89.9%	1.705	88.7%	87.3%
	SOFTS	88.8%	1.624	87.3%	86.6%	90.9%	1.766	89.0%	89.4%	90.5%	1.879	87.4%	89.6%	91.3%	1.791	89.6%	89.5%
	ave	89.2%	1.683	87.5%	87.5%	90.2%	1.742	88.3%	88.3%	90.5%	1.844	87.6%	89.3%	90.3%	1.748	88.1%	87.8%
ETTth2	itransformer	87.9%	1.336	85.7%	86.9%	89.7%	1.401	86.3%	86.2%	88.0%	1.258	84.7%	87.2%	90.4%	1.416	85.6%	85.4%
	leddam	86.7%	1.133	82.8%	85.4%	87.0%	1.141	82.8%	85.4%	88.1%	1.130	82.9%	86.1%	86.8%	1.119	79.0%	83.5%
	SOFTS	87.1%	1.113	82.7%	85.9%	87.1%	1.105	82.2%	85.9%	87.1%	1.103	83.2%	86.2%	86.5%	1.073	77.4%	84.3%
	ave	87.3%	1.194	83.8%	86.0%	88.0%	1.216	83.8%	85.8%	87.7%	1.164	83.6%	86.5%	87.9%	1.203	80.7%	84.4%
PEMS03	itransformer	88.9%	1.139	83.5%	88.3%	89.3%	1.154	82.3%	88.7%	89.3%	1.163	84.0%	88.8%	89.5%	1.140	63.5%	87.2%
	leddam	88.4%	1.125	77.9%	87.7%	88.5%	1.124	75.3%	87.5%	88.3%	1.122	78.0%	87.7%	87.2%	1.063	52.8%	83.0%
	SOFTS	88.9%	1.274	84.5%	88.0%	90.0%	1.308	85.5%	89.4%	88.6%	1.269	84.3%	88.0%	90.9%	1.326	79.8%	88.7%
	ave	88.7%	1.179	82.0%	88.0%	89.3%	1.195	81.0%	88.6%	88.8%	1.185	82.1%	88.2%	89.2%	1.176	65.3%	86.3%
PEMS04	itransformer	89.4%	1.146	81.6%	88.9%	90.3%	1.197	82.3%	89.7%	89.1%	1.124	81.3%	88.8%	91.9%	1.236	64.2%	90.9%
	leddam	88.6%	1.083	83.3%	88.0%	89.3%	1.106	84.3%	88.7%	88.4%	1.071	83.7%	87.8%	89.5%	1.093	76.7%	87.4%
	SOFTS	88.8%	1.359	81.0%	87.9%	89.2%	1.366	82.8%	88.3%	88.8%	1.334	80.8%	87.8%	89.4%	1.354	79.6%	87.1%
	ave	88.9%	1.196	82.0%	88.3%	89.6%	1.223	83.1%	88.9%	88.8%	1.176	81.9%	88.1%	90.3%	1.227	73.5%	88.5%
PEMS07	itransformer	89.9%	1.048	80.1%	89.7%	90.5%	1.032	75.2%	90.1%	89.5%	1.001	75.9%	89.2%	93.4%	1.103	48.2%	92.2%
	leddam	89.6%	1.005	77.9%	89.3%	90.1%	1.020	75.0%	89.8%	89.5%	1.008	77.8%	89.1%	92.0%	1.058	49.7%	90.7%
	SOFTS	89.5%	1.171	83.6%	89.1%	90.5%	1.205	84.5%	90.1%	89.2%	1.172	83.2%	88.8%	92.9%	1.257	74.5%	91.6%
	ave	89.6%	1.075	80.5%	89.4%	90.4%	1.086	78.2%	90.0%	89.4%	1.060	79.0%	89.0%	92.8%	1.139	57.5%	91.5%
PEMS08	itransformer	89.1%	1.094	80.3%	88.6%	89.7%	1.127	79.6%	88.8%	89.1%	1.103	80.3%	88.6%	90.3%	1.128	69.4%	88.0%
	leddam	88.3%	1.100	76.2%	87.6%	89.0%	1.120	75.0%	87.8%	88.2%	1.101	76.2%	87.5%	88.7%	1.090	64.6%	85.7%
	SOFTS	88.6%	1.235	76.3%	87.8%	89.6%	1.260	74.3%	88.5%	88.2%	1.205	74.8%	87.7%	89.9%	1.243	67.9%	86.9%
	ave	88.7%	1.143	77.6%	88.0%	89.4%	1.169	76.3%	88.4%	88.5%	1.137	77.1%	87.9%	89.6%	1.154	67.3%	86.9%
1st Count		12				3				20				12			

Overall, employing LSTM as the error quantile prediction model leads to modest improvements over the original method in many scenarios. Across our comprehensive evaluation of 48 test cases, it demonstrated particularly competitive performance, achieving optimal results in 20 instances. However, it is important to note that these improvements remain relatively minor in magnitude.

The other two extension methods - the ECI-based interval length adjustment and the adaptive learning rate approach - showed limited effectiveness in enhancing the original performance. In fact, these modifications frequently introduced negative effects.

Limitations and broader impact

We acknowledge that our method has several limitations. The features obtained from the deep learning model are designed for predicting mean values rather than quantiles, which means their effectiveness in predicting error quantiles may be constrained. Although our experimental results demonstrate that these features still retain some predictive value and can provide meaningful information about error quantiles, future improvements could consider enhancing feature utilization efficiency. Alternatively, during the pre-training phase of point prediction models, methods to improve feature expressiveness might be explored - such as incorporating causal inference approaches to help the features capture truly influential factors underlying the time series patterns.

Second, the learning rate remains a crucial hyperparameter in our framework. Our sensitivity analysis revealed that overly small learning rates can lead to inadequate coverage probability, necessitating rate adjustment. This highlights the open challenge of determining optimal learning rates - an interesting research question and a limitation of our current approach. Future work could explore adaptive learning rate strategy as a viable solution to this challenge.

Finally, our current experiments are based on conventional deep learning architectures, recent advances have successfully applied large models to time series forecasting. This naturally raises an important research direction for future work - investigating the compatibility of our approach when integrated with large foundation models for time series prediction.

Regarding broader societal impacts, the methodology proposed in this study can provide reliable time-series predictions for critical sectors including finance, healthcare, and environmental monitoring, thereby enhancing decision-making quality. However, it must be emphasized that the effectiveness of our approach is fundamentally contingent upon the quality of training data and point prediction model.

Visualization of confidence intervals

We plot the predicted confidence intervals based on SOFTS (Han et al. 2024) in different dataset in the following figures. In these figures, the right subfigure in the lower row represent our method. Other subfigures represent TQA-E, ACI, ECI, LPCI and PIC, respectively.

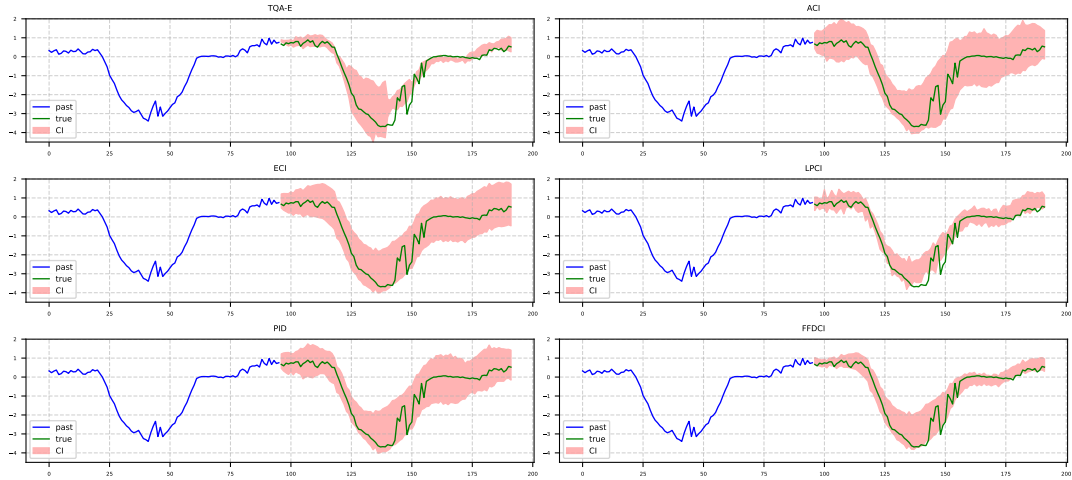


Figure 4: Predicted confidence intervals in ETTm1 dataset

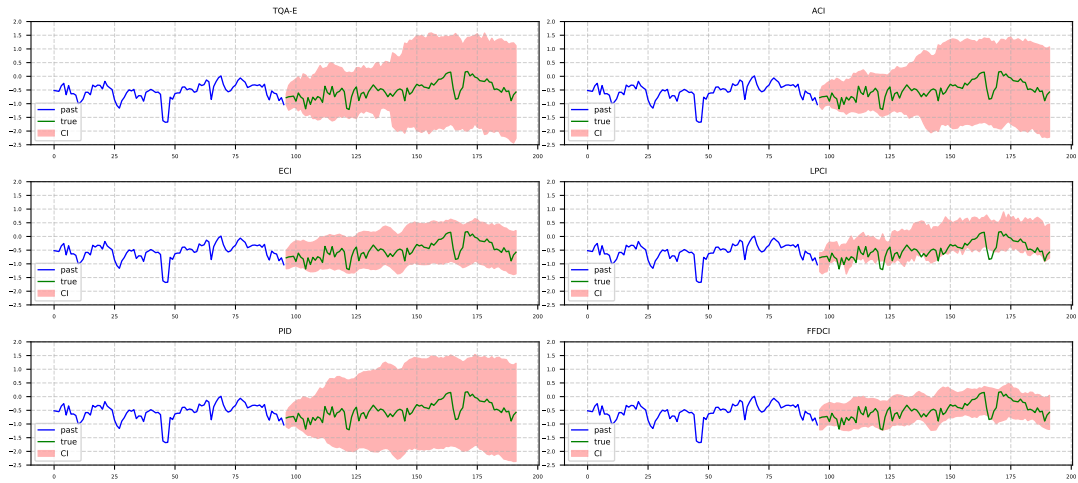


Figure 5: Predicted confidence intervals in ETTm2 dataset



Figure 6: Predicted confidence intervals in ETTh1 dataset

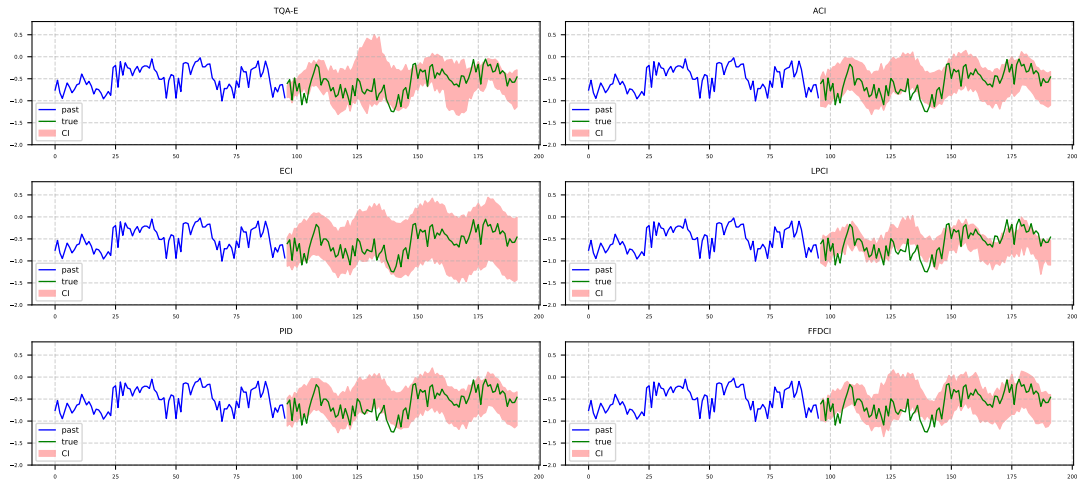


Figure 7: Predicted confidence intervals in ETTh2 dataset

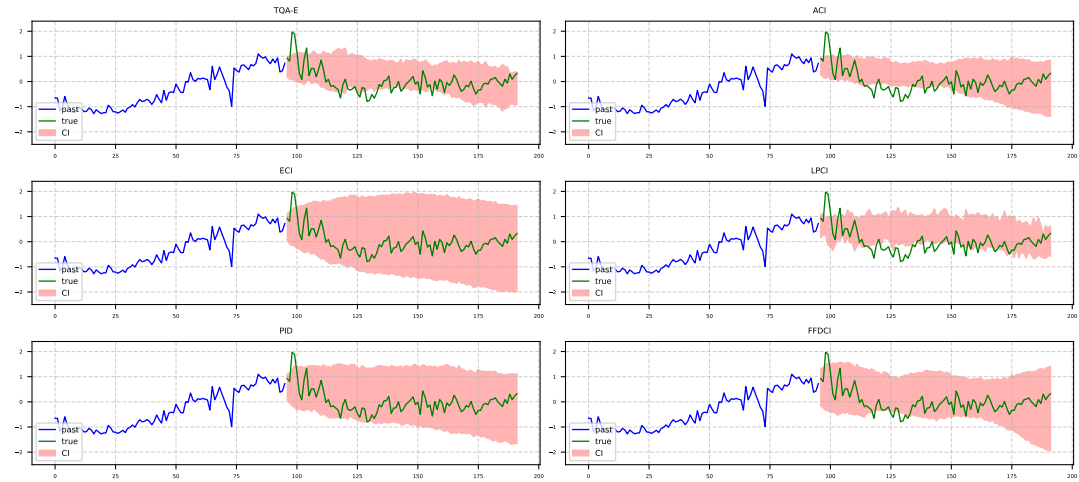


Figure 8: Predicted confidence intervals in weather dataset

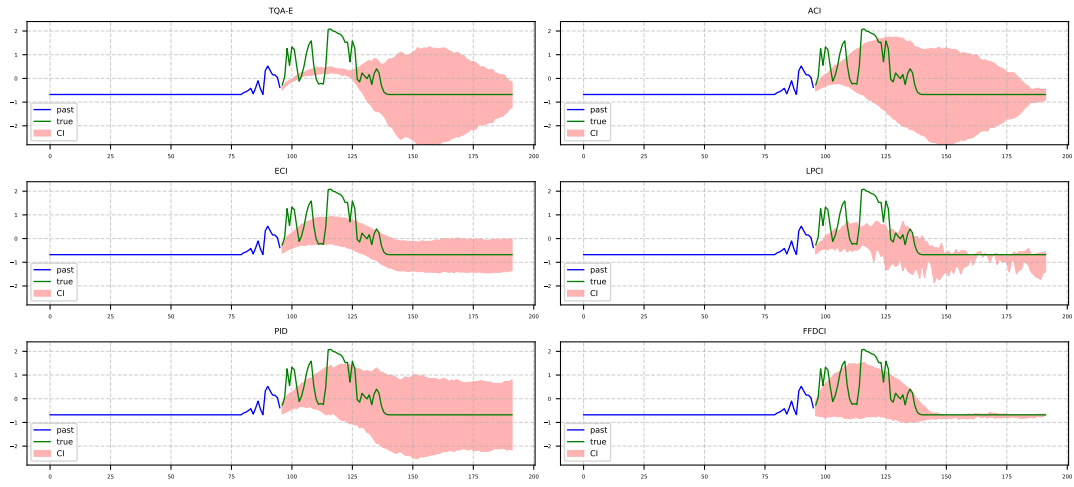


Figure 9: Predicted confidence intervals in solar dataset

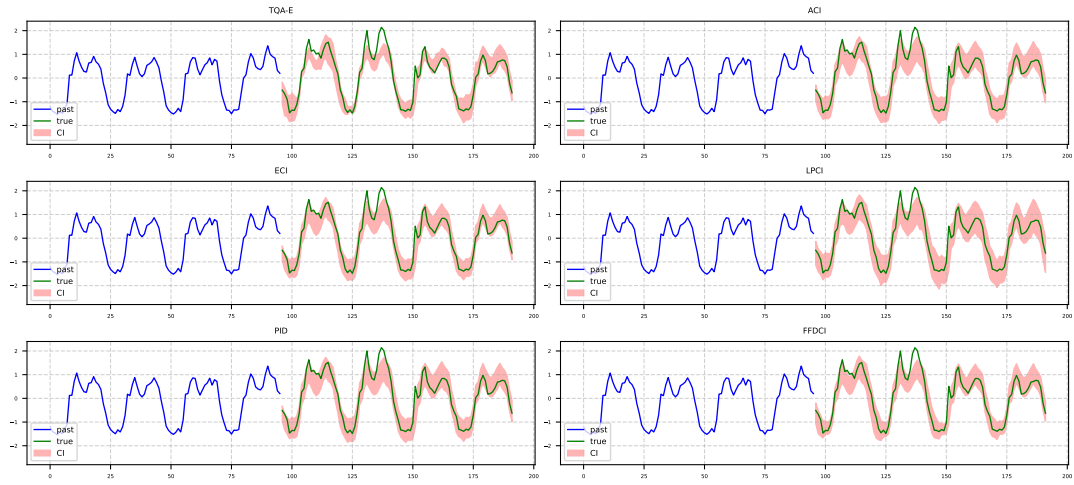


Figure 10: Predicted confidence intervals in electricity dataset

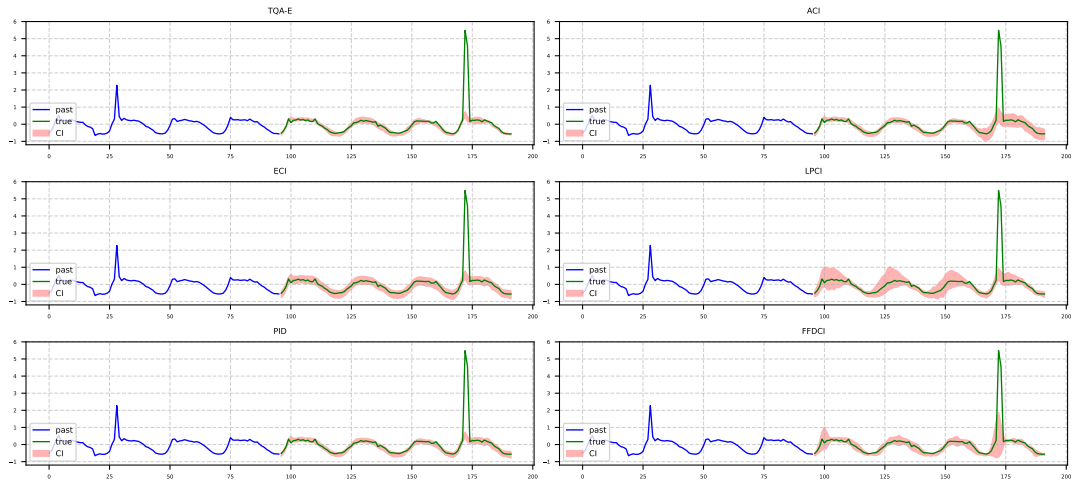


Figure 11: Predicted confidence intervals in traffic dataset

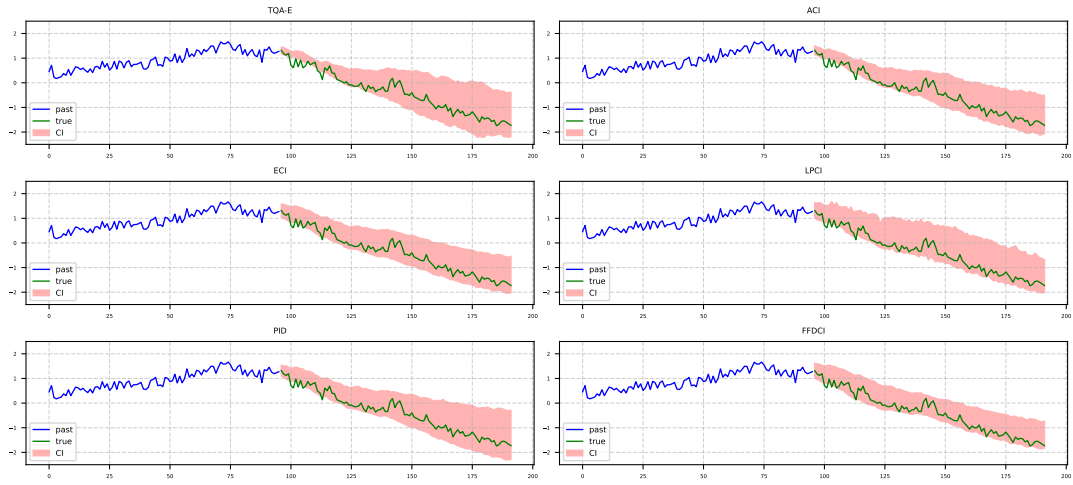


Figure 12: Predicted confidence intervals in PEMS03 dataset

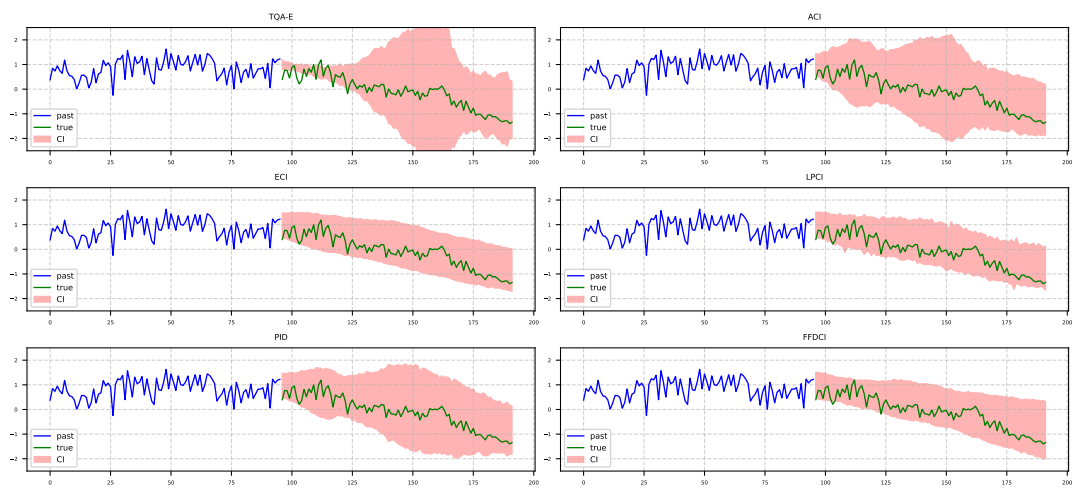


Figure 13: Predicted confidence intervals in PEMS04 dataset

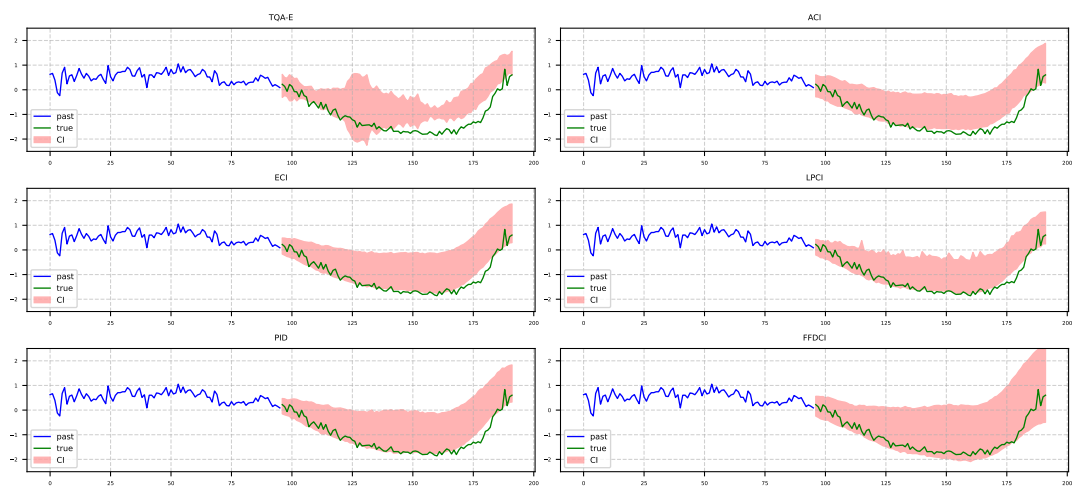


Figure 14: Predicted confidence intervals in PEMS07 dataset

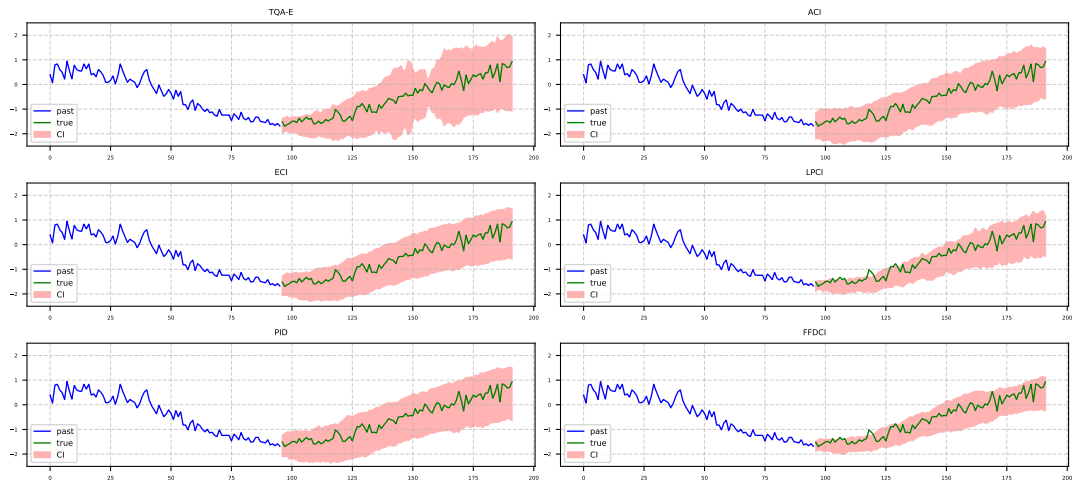


Figure 15: Predicted confidence intervals in PEMS08 dataset